

RIVER WATER QUALITY IN HONG KONG IN 2019



Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Mission

To conduct a comprehensive and scientific monitoring programme that helps safeguard the health of Hong Kong's rivers and streams and achieve the Water Quality Objectives.



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Acknowledgement

We acknowledge the Government Laboratory for undertaking chemical analyses of the water samples.

Abbreviations / 簡稱

Ammonia Nitrogen	NH ₄ -N	氨氮
Chemical Oxygen Demand	COD	化學需氧量
Dissolved Oxygen	DO	溶解氧
Dry Weather Flow Interceptors	DWFI	旱季截流器
Environmental Protection Department	EPD / 環保署	環境保護署
<i>Escherichia coli</i>	<i>E. coli</i>	大腸桿菌
Livestock Waste Control Scheme	LWCS	禽畜廢物管制計劃
Sewerage Master Plans	SMPs	污水收集整體計劃
Suspended Solids	SS	懸浮固體
Water Control Zone	WCZ	水質管制區
Water Pollution Control Ordinance	WPCO	水污染管制條例
Water Quality Index	WQI	水質指數
Water Quality Objectives	WQO	水質指標
5-day Biochemical Oxygen Demand	BOD ₅	五天生化需氧量

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1. Introduction

The primary beneficial uses of Hong Kong's rivers and streams, generally short with small flows, are for supporting aquatic life, general amenity, storm water drainage and flood prevention. Many of the watercourses however have upstream sections located within the catchment areas of reservoirs and the base flow is largely extracted for supply of potable water.

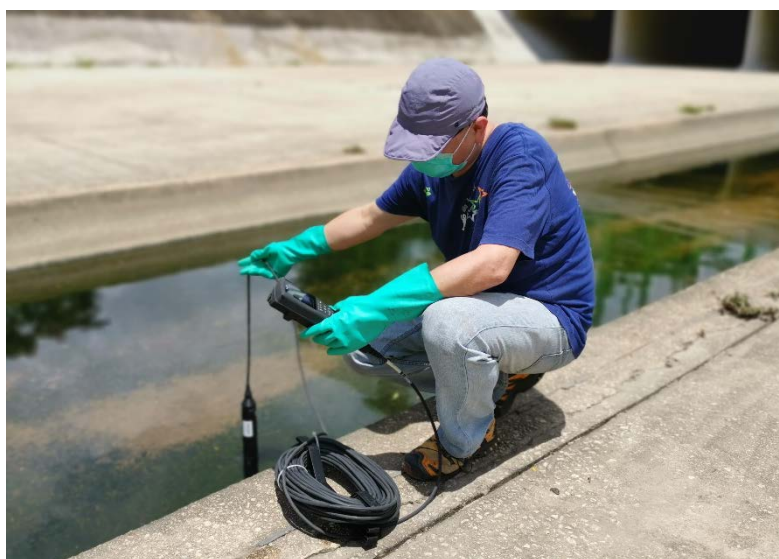
The Environmental Protection Department (EPD) has implemented a comprehensive river water quality monitoring programme since 1986 to provide data for water quality management and pollution control purposes. One or more representative monitoring stations have been set up at the upstream and downstream sections of the main channels and tributaries of large rivers in the New Territories or those flowing through the urban areas. A number of small rivers and streams in rural areas and outlying islands are also monitored.

The monitoring programme serves the following purposes:

- evaluate the pollution status of rivers;
- monitor long-term changes in river water quality;
- provide scientific basis for planning water pollution control strategies;
- assess compliance with the Water Quality Objectives (WQOs); and
- compile Water Quality Index (WQI) to reflect the overall state and trend of the health of rivers.

This report summarises the water quality of rivers and streams covered by EPD's river monitoring programme in 2019. Annual reports can be downloaded from the EPD's website:

<https://www.epd.gov.hk/epd/english/environmentinhk/water/hkwqrc/waterquality/river-2.html>



In situ water quality measurements at a tributary of Shing Mun River

2. An Overview of Hong Kong's River Water Quality in 2019

In 2019, the EPD's river monitoring programme covered 30 watercourses and a total of 82 monitoring stations (Appendix A), as compared with some 14 watercourses and 47 stations when the programme started in 1986. The monitoring programme involved monthly field measurements and water sample collection for laboratory analyses. Over 50 physico-chemical and biological parameters, including organic matters, nutrients, metals and *E. coli*, were analysed (Appendix B).

2.1. Compliance Rate of Water Quality Objectives (WQOs)

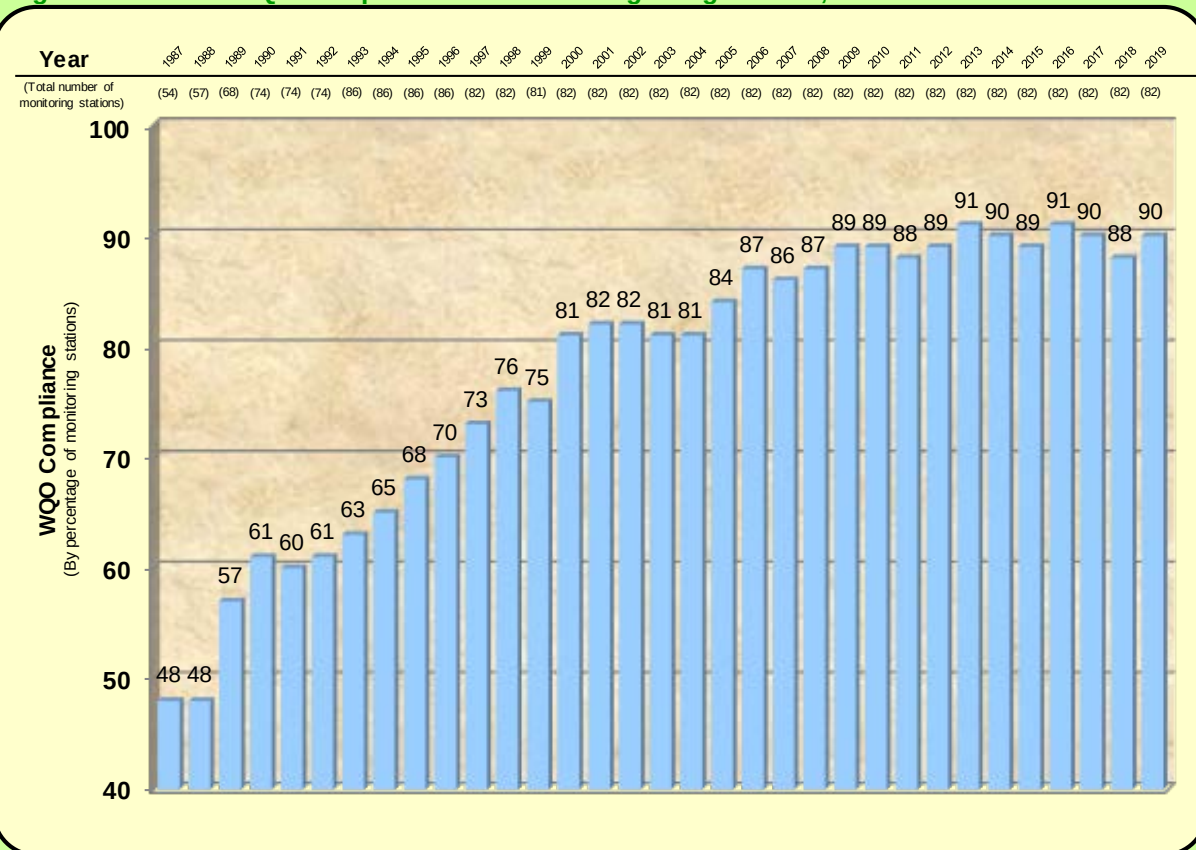
Five representative parameters, including pH, suspended solids (SS), dissolved oxygen (DO), 5-day biochemical oxygen demand (BOD₅) and chemical oxygen demand (COD), are used to assess compliance with the WQOs applicable for individual monitoring stations (Appendix C). This report presents the annual average compliance rates for individual watercourses as well as the overall compliance rate for the rivers and streams monitored in Hong Kong (Appendix F).

For rivers that have more than one monitoring stations, the WQO compliance rate was expressed as an average of the annual compliance rates of all their individual monitoring stations. Similarly, the overall compliance rate for the entire Hong Kong's river water quality was the overall average rate for all the monitoring stations in the territory. The river water quality in Hong Kong in 2019 continued to be satisfactory with a high overall compliance rate of 90% which is within the range of fluctuations over the past ten years (88% – 91%) (Figure 1). Figure 2 shows the WQO compliance rates for key river water quality parameters between 1987 and 2019.

High WQO compliance rates in recent years have been achieved owing to the implementation of pollution control measures, including continuous extension of the sewer networks to more villages under the Sewerage Master Plans (SMPs) as well as enforcement of the Water Pollution Control Ordinance and the Livestock Waste Control Scheme introduced under the Waste Disposal Ordinance.

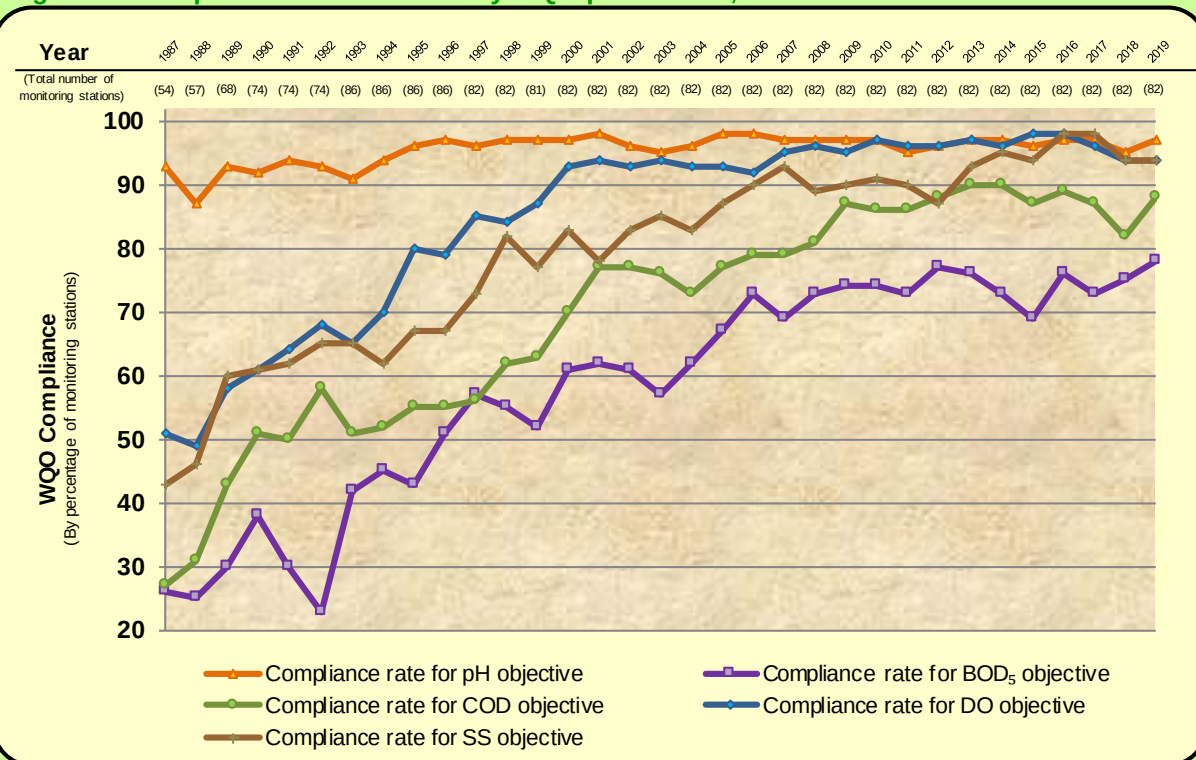
In summary, the state of water quality of Hong Kong's watercourses remained good and steady in 2019 and followed a long-term improvement trend as a result of the gradual reduction of pollution loading to the rivers and streams.

Figure 1. Overall WQO compliance rates for Hong Kong's rivers, 1987-2019



Figures are rounded to the nearest integer

Figure 2. Compliance rates for five key WQO parameters, 1987-2019



2.2. Water Quality Index (WQI) Gradings

WQI is relevant to conserving the primary beneficial use of rivers and streams for supporting aquatic life. It rates the overall state of health of rivers and streams into five gradings, namely “Excellent”, “Good”, “Fair”, “Bad” and “Very Bad”, by assessing three key water quality parameters: DO, BOD₅ and ammonia-nitrogen levels¹.

In 2019, 85% of the river monitoring stations were graded “Good” or “Excellent”, as compared with only 26% in 1987 (Figure 3), suggesting that the river water quality has greatly improved and the pollution loadings in these watercourses have been substantially reduced in the past three decades. Majority of the river monitoring stations located in the Eastern New Territories, Southwestern New Territories, Lantau Island and Kowloon fell into these two WQI gradings. For comparison, only 6% of the monitoring stations were graded “Bad” and none classified as “Very Bad” in 2019, while 54% stations in these two gradings were recorded in 1987. Most of the “Bad” stations are located along watercourses in the Northwestern New Territories (Figures 4 and 5). Among the 82 monitoring stations, minor changes in WQI gradings were observed in 2019 for 12 monitoring stations, which are generally within the normal range of natural fluctuations over the past ten years (Table 1).

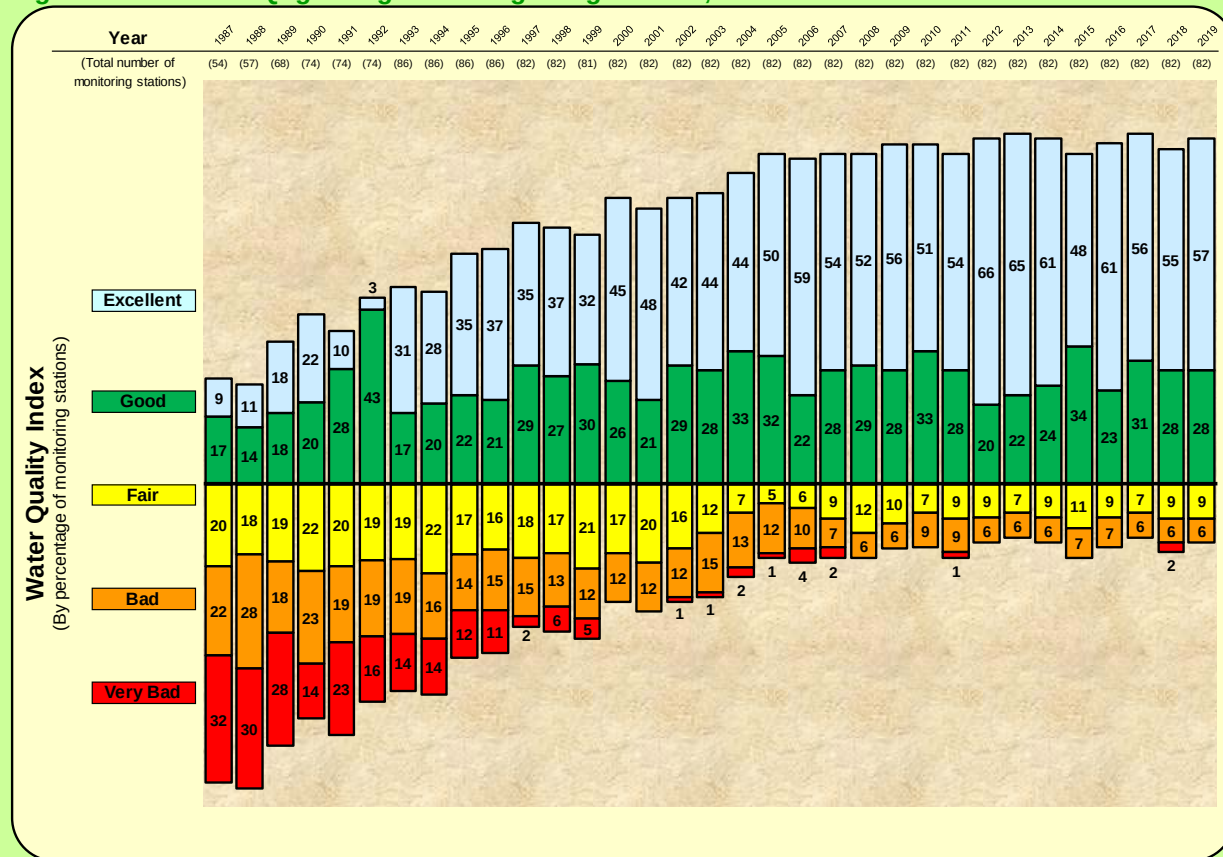
Table 1. River monitoring stations with changes in WQI gradings (2018 - 2019)

River monitoring station	WQI grading	
	2018	2019
Shing Mun River (TR17L)	Good	Excellent
Shing Mun River (TR19)	Excellent	Good
Shing Mun River (TR19I)	Excellent	Good
Tseng Lan Shue Stream (JR3)	Bad	Fair
Tseng Lan Shue Stream (JR6)	Good	Excellent
River Indus (IN1)	Fair	Good
River Beas (RB1)	Good	Excellent
River Beas (RB3)	Fair	Good
Yuen Long Creek (YL2)	Bad	Fair
Yuen Long Creek (YL3)	Very Bad	Bad

¹ Please refer to Appendix G for more details on the calculation and assessment of WQI.

Yuen Long Creek (YL4)	Very Bad	Bad
Mui Wo River (MW5)	Good	Excellent

Figure 3. Overall WQI gradings for Hong Kong's rivers, 1987-2019



Figures are rounded to the nearest integer

Figure 4. Map of river monitoring stations and their WQI gradings in 2019



Figure 5. Map of river monitoring stations and their WQI gradings in 1987

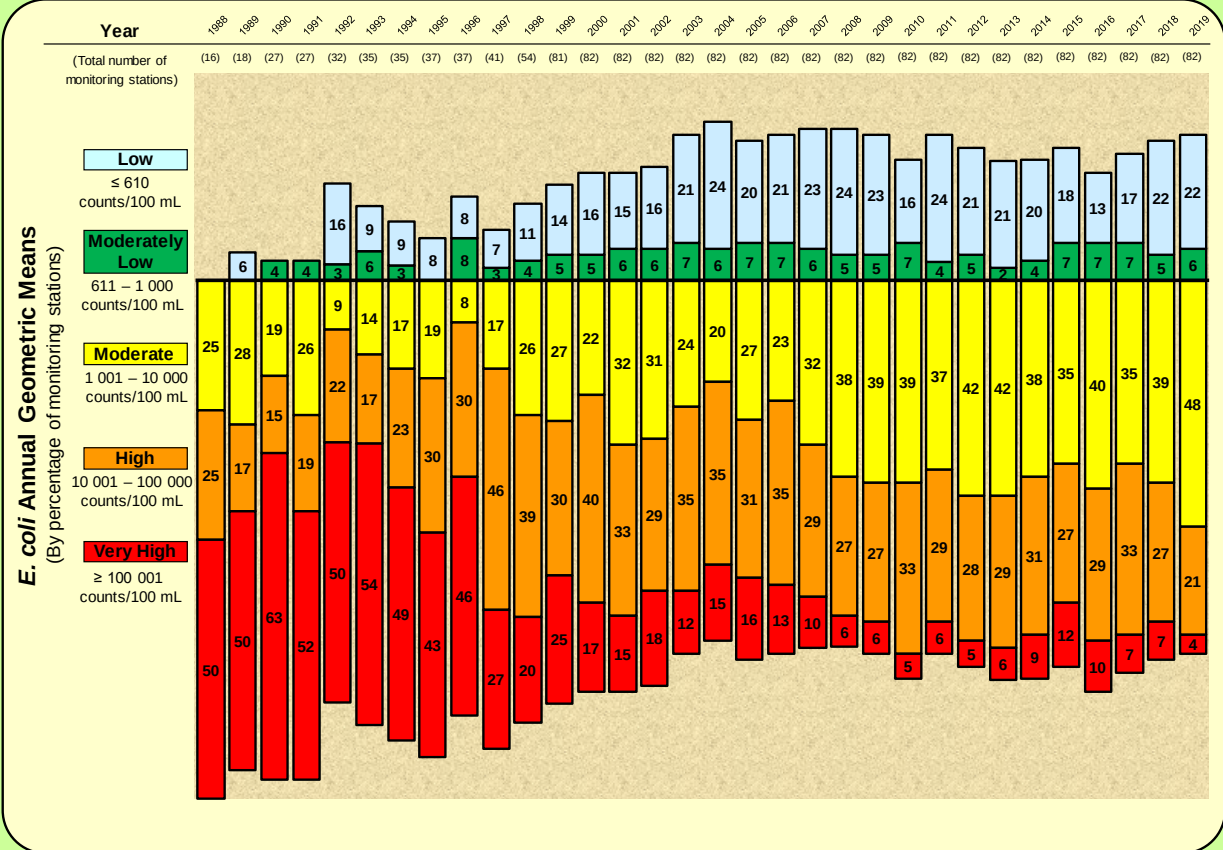


As faeces from all warm-blooded animals (including humans, livestock animals, pets, birds, etc.) contain *E. coli*, the measured level in a water body has been widely used as an indicator for assessing the presence and extent of faecal contamination. In 2019, 28% of the monitoring stations had “Low” or “Moderately Low” levels of *E. coli* (i.e., not exceeding 1 000 counts/100mL) while 25% stations recorded “High” or “Very High” *E. coli* levels² (i.e., over 10 000 counts/100mL) (Figure 6).

Most of the monitoring stations with “Very High” *E. coli* levels are located in the northwest part of the New Territories (Figure 7), mainly due to discharges from livestock farms, surface runoffs from unsewered villages as well as expedient connections in old districts. While these watercourses are mainly used for storm water conveyance and flood prevention, the Government will continue its efforts in enforcement of pollution control legislations and extension of the public sewer networks to unsewered villages so as to further reduce pollution at these locations.

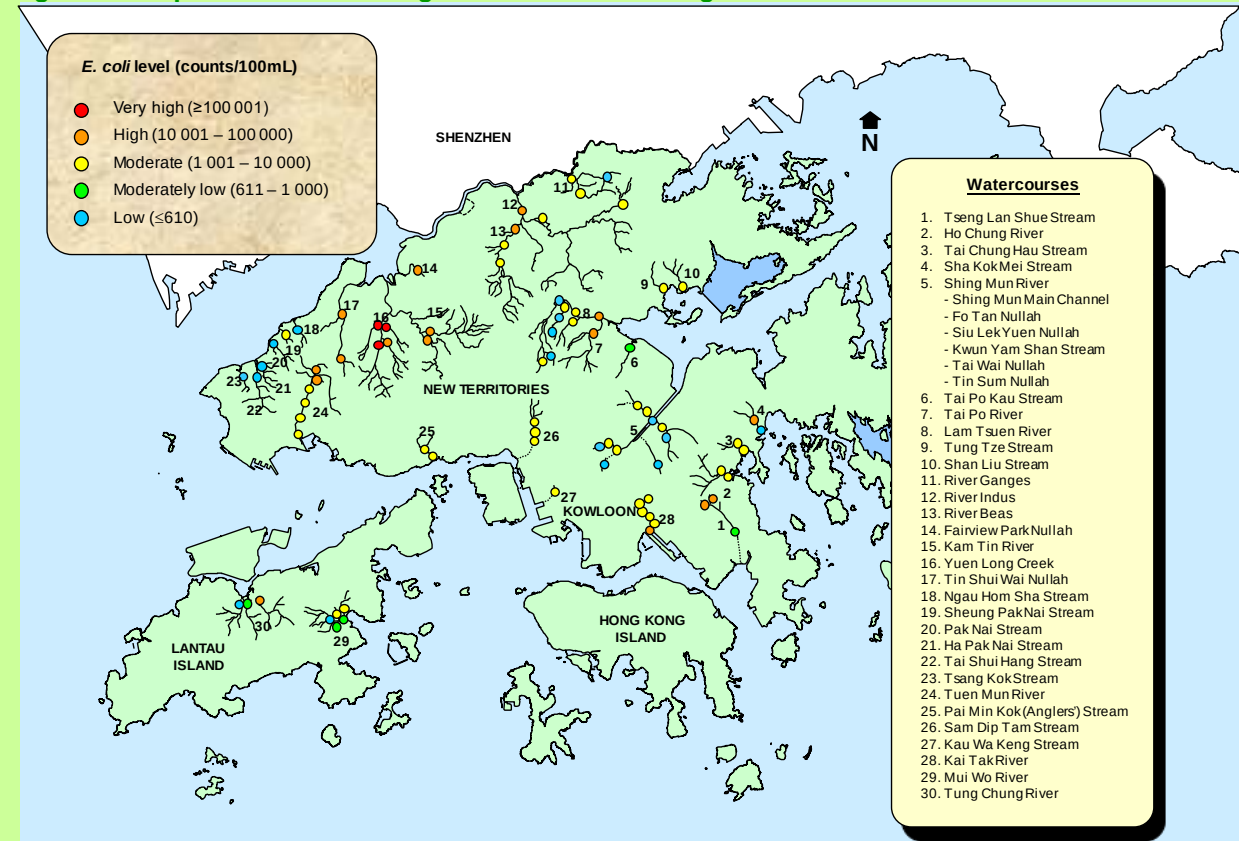
² All levels of *E. coli* in this report are reported as annual geometric means (counts/100mL).

Figure 6. Overall annual geometric mean *E. coli* levels in Hong Kong's rivers (1988-2019)



Figures are rounded to the nearest integer

Figure 7. Map of river monitoring stations and annual geometric mean *E. coli* levels in 2019



3. Water Quality of Rivers and Streams in Various Districts

3.1. Eastern New Territories

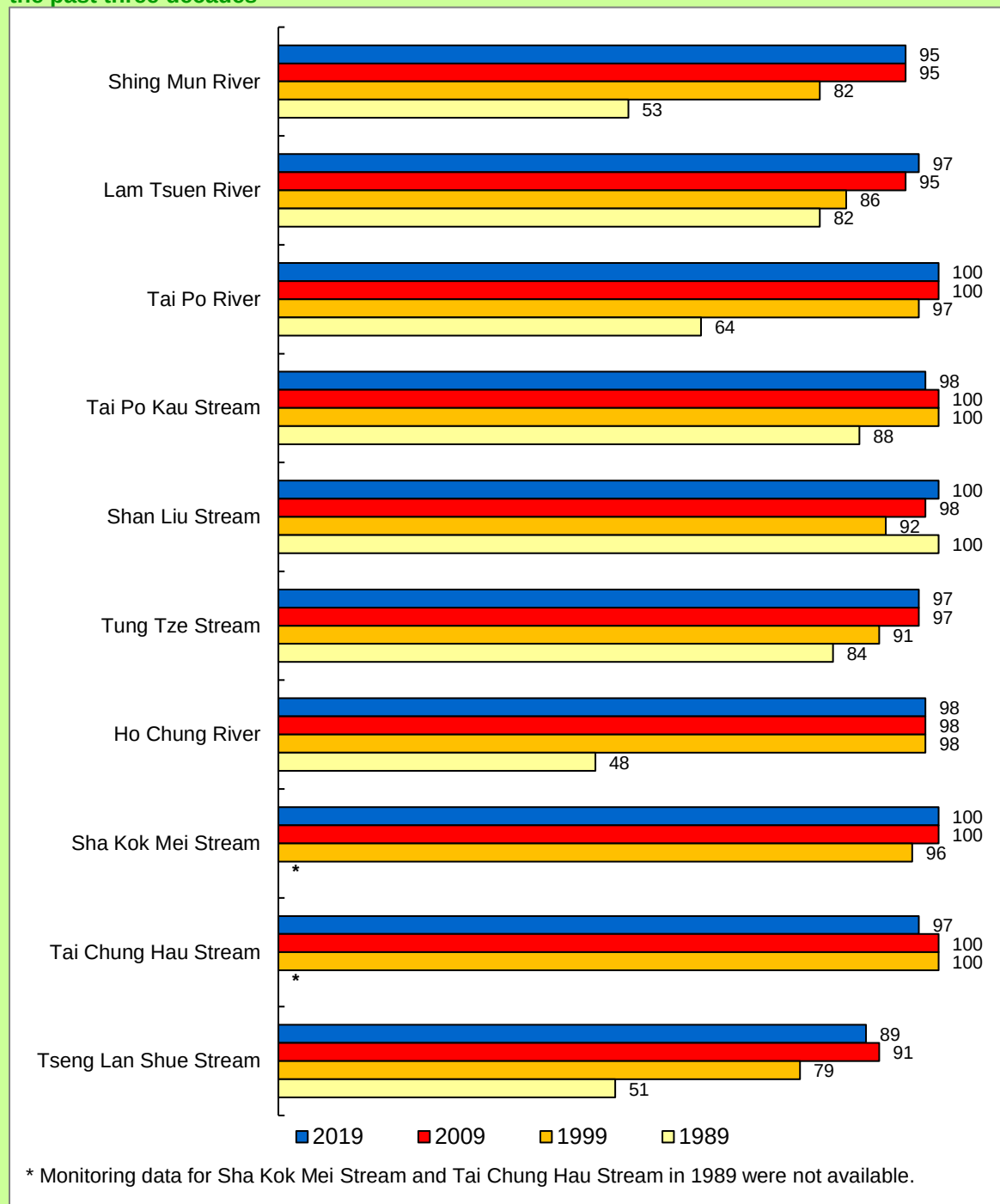
The EPD monitored ten watercourses in the Eastern New Territories, six of which, i.e., Shing Mun River, Lam Tsuen River, Tai Po River, Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream, are located in the Tolo Harbour and Channel Water Control Zone (WCZ). Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream are within the Port Shelter WCZ, while Tseng Lan Shue Stream belongs to the Junk Bay WCZ.

The water quality of rivers and streams in the Eastern New Territories is generally good. In 2019, the overall WQO compliance rate of all river monitoring stations in these areas remained high at 96%, as compared with 96% in 2009, 88% in 1999 and 66% in 1989. Three pristine rivers within these areas, including Tai Po River, Shan Liu Stream and Sha Kok Mei River, all achieved the WQOs in 2019 (Figure 8).



Shing Mun River

Figure 8. WQO compliance rates (%) for rivers and streams in Eastern New Territories over the past three decades



Being the largest river in Hong Kong, Shing Mun River has three main tributaries and runs through the densely populated Sha Tin urban area. The river has shown marked improvement over the past three decades, and maintained a satisfactory WQO compliance rate of 95% in 2019. The rivers in the Tai Po District also achieved high WQO compliance in 2019, while Lam Tsuen River, the major river draining through the urban area of Tai Po and joining Tai Po River before entering Tolo Harbour, achieved 97% compliance rate. Similarly, the compliance rates for Tai Po Kau Stream and Tung Tze Stream also reached 98% and 97% respectively.



Lam Tsuen River

In the Port Shelter WCZ, Ho Chung River and Tai Chung Hau Stream achieved high WQO compliances of 98% and 97% respectively in 2019.

Tseng Lan Shue Stream in the Junk Bay WCZ had a WQO compliance rate of 89% in 2019.

As for the WQI gradings, 31 out of the 32 river monitoring stations in the Eastern New Territories (or 97%) were rated as “Good” or “Excellent” in 2019, same as the situation in 2018 (Figures 9 to 13). Only the monitoring station (JR3) located at the Tseng Lan Shue Stream was graded “Fair” (Figure 12).

Figure 9. WQI gradings and *E. coli* levels in Shing Mun River

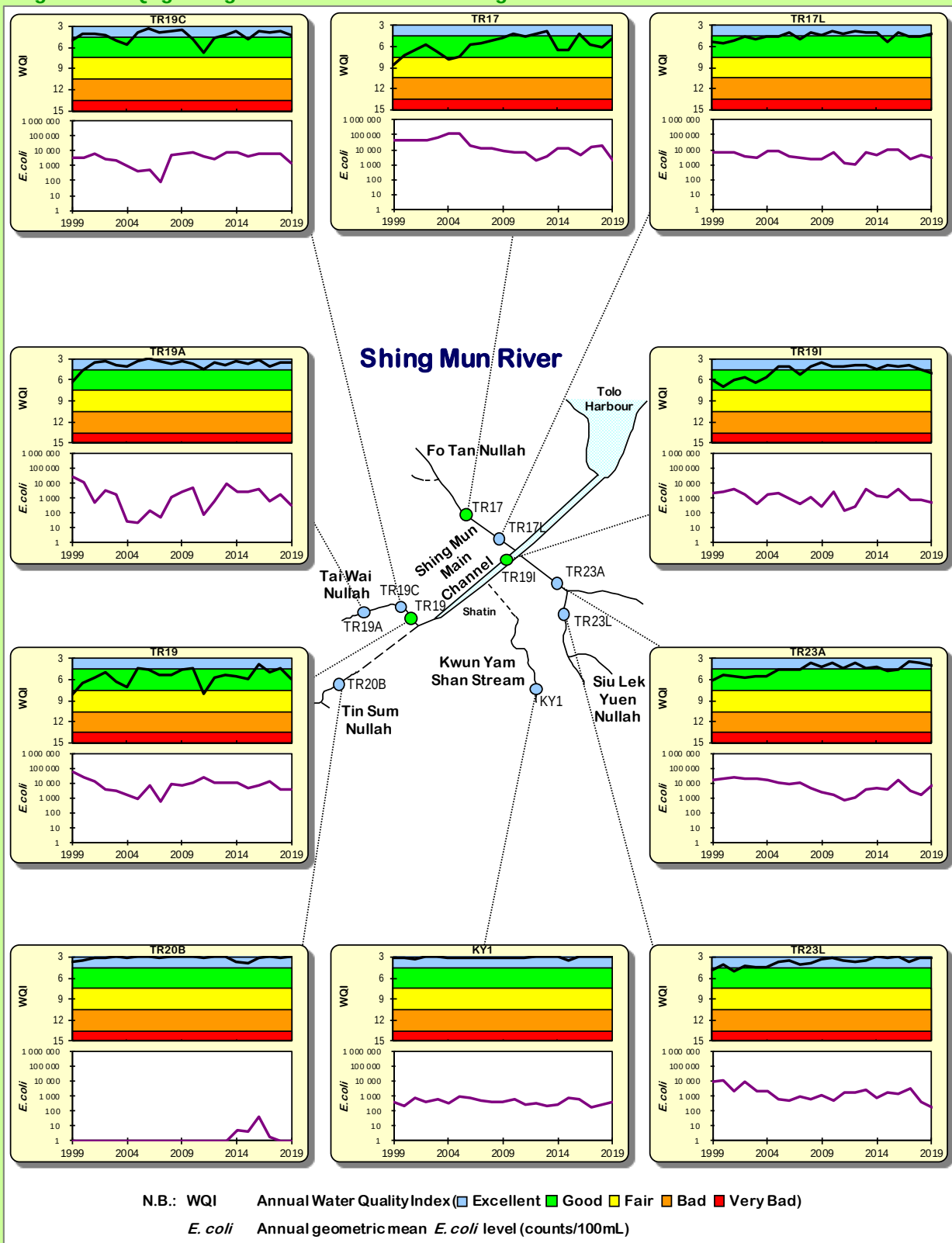


Figure 10. WQI gradings and *E. coli* levels in Lam Tsuen River and Tai Po River

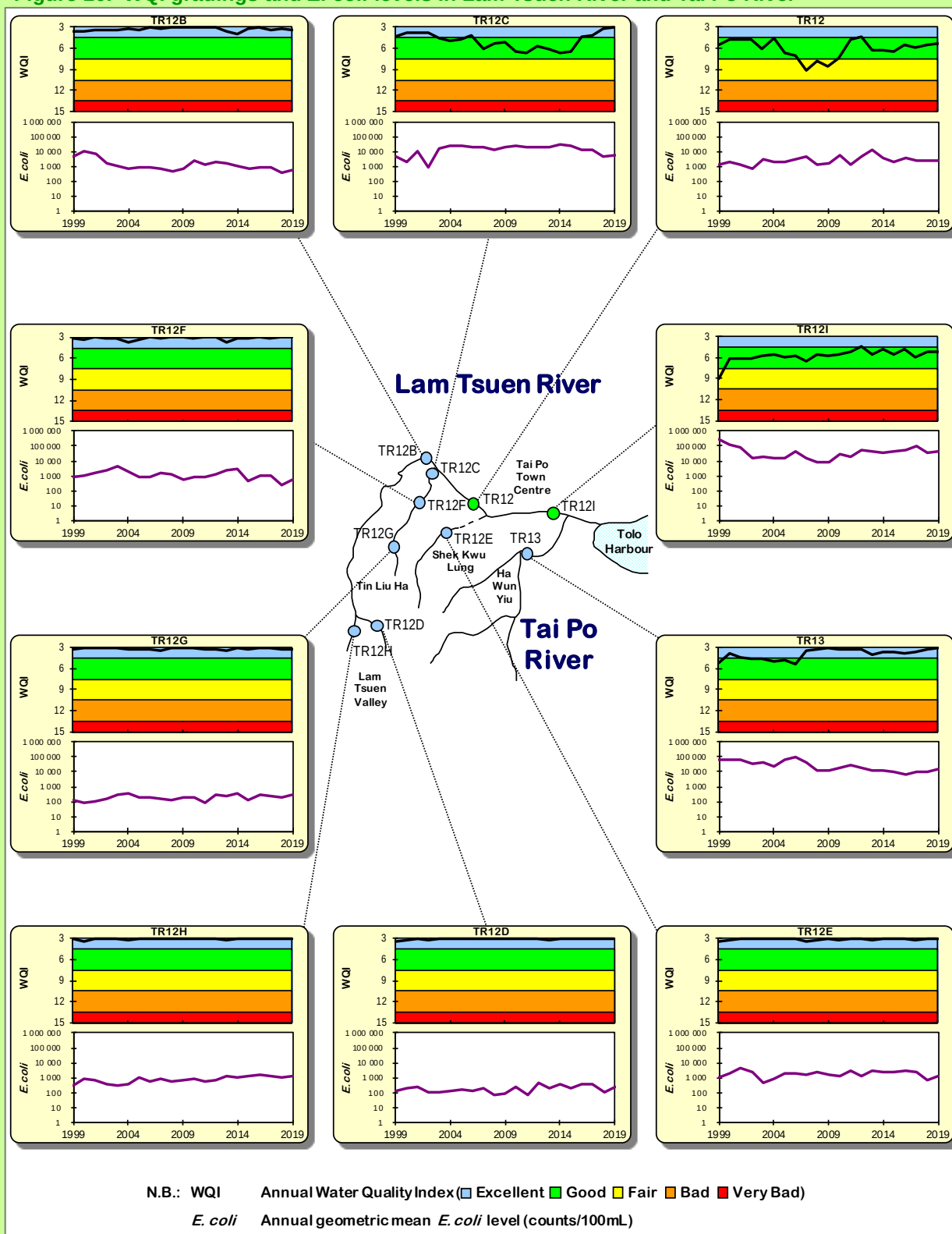


Figure 11. WQI gradings and *E. coli* levels in Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream

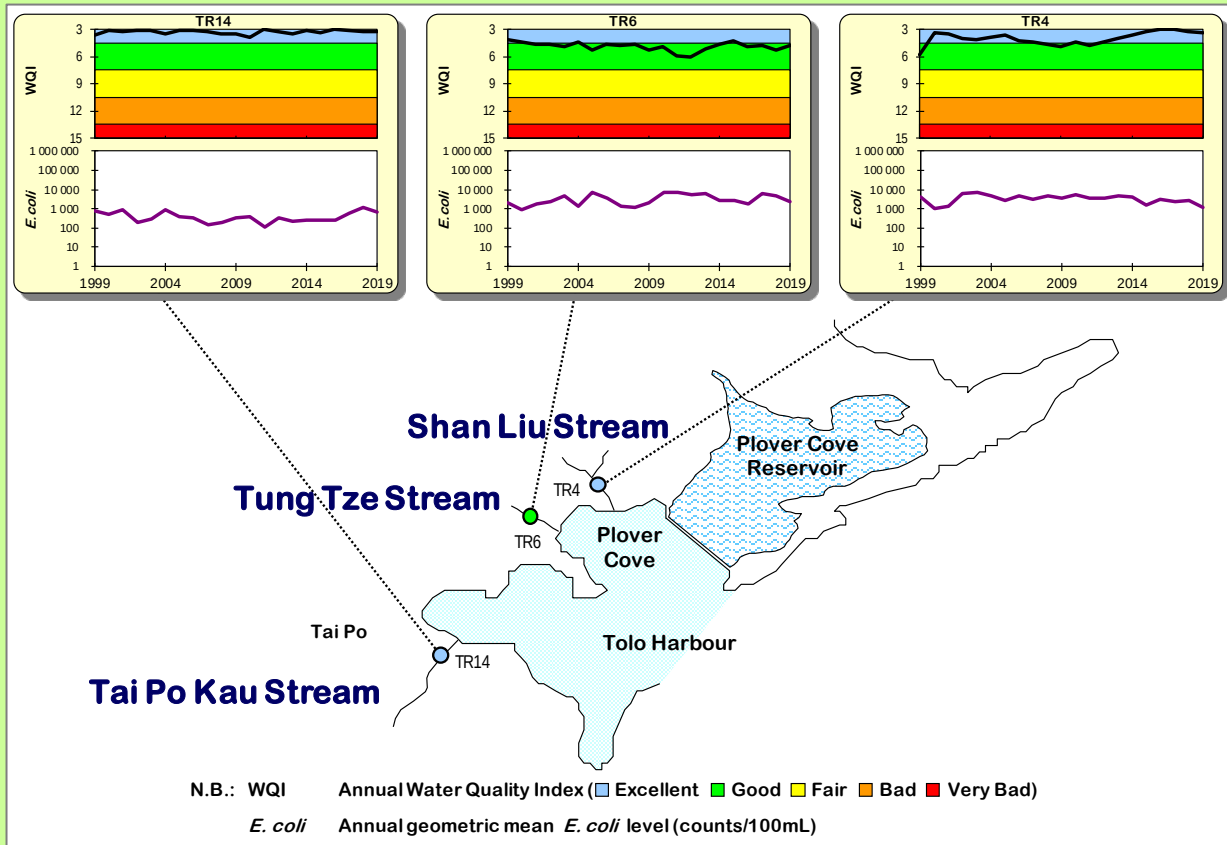


Figure 12. WQI gradings and *E. coli* levels in Tseng Lan Shue Stream

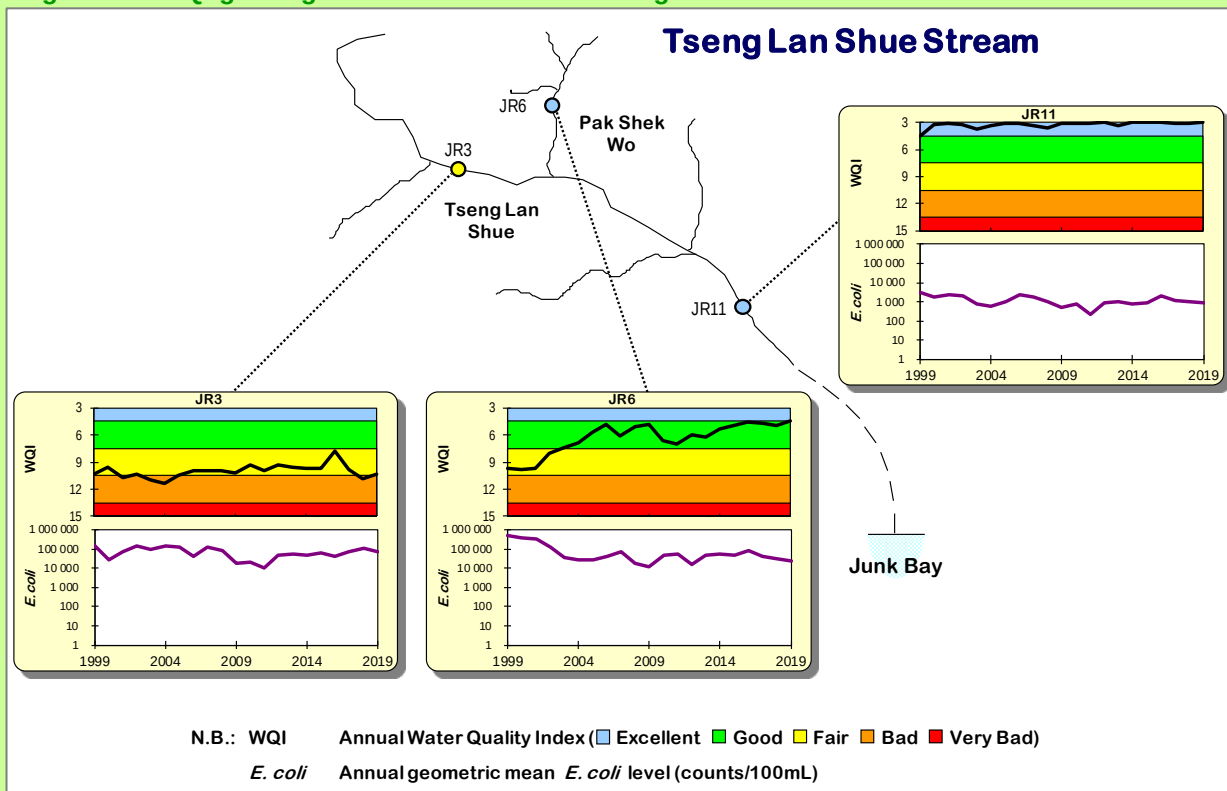
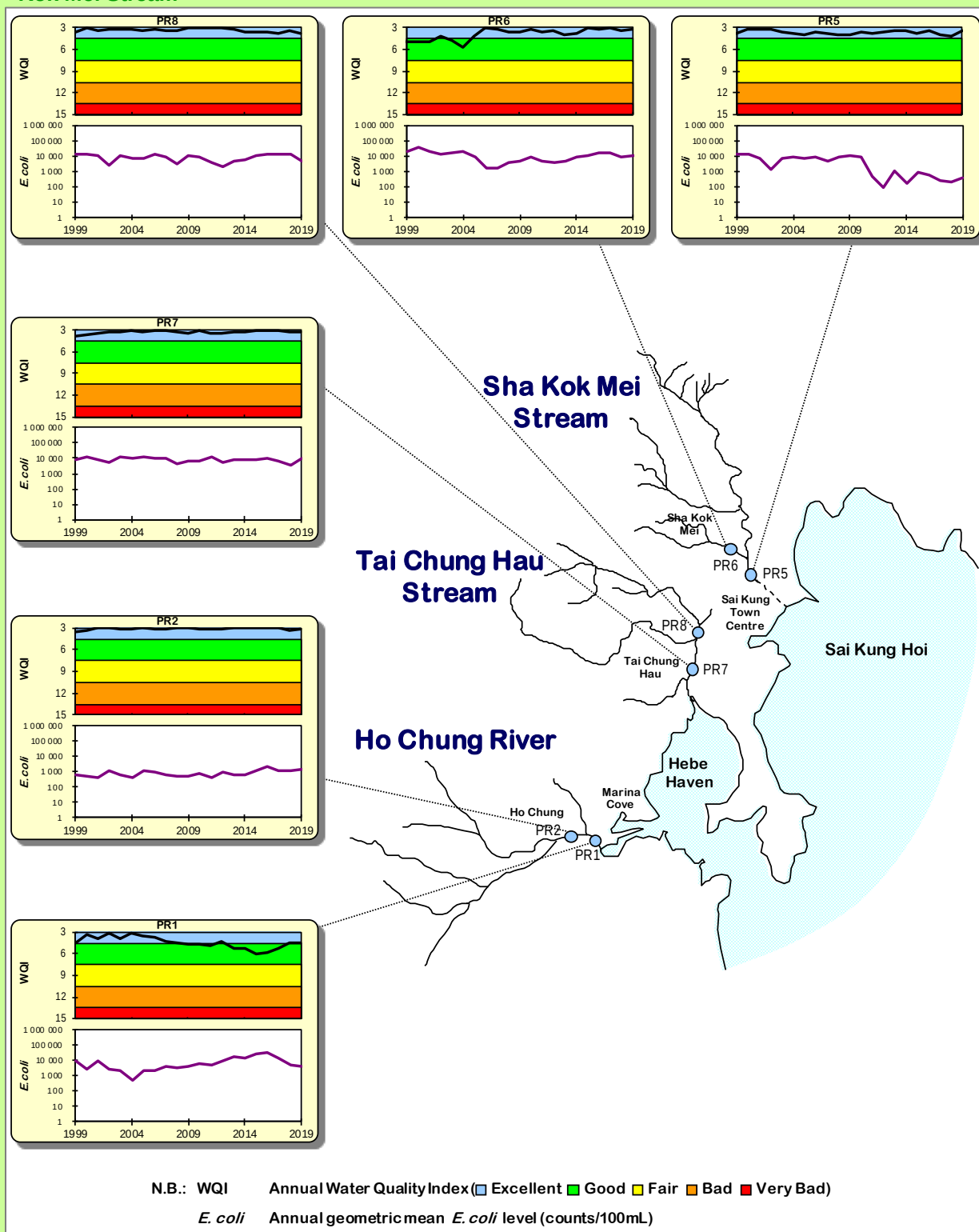


Figure 13. WQI gradings and *E. coli* levels in Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream



3.2. Northwestern New Territories

The EPD monitors, within the Deep Bay WCZ, a total of 13 rivers and streams draining into Shenzhen River or directly into Deep Bay (Shenzhen Bay). These include River Indus, Beas and Ganges in the North District; Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah in the Yuen Long District, and as well as 6 smaller streams located around Lau Fau Shan area.



River Beas

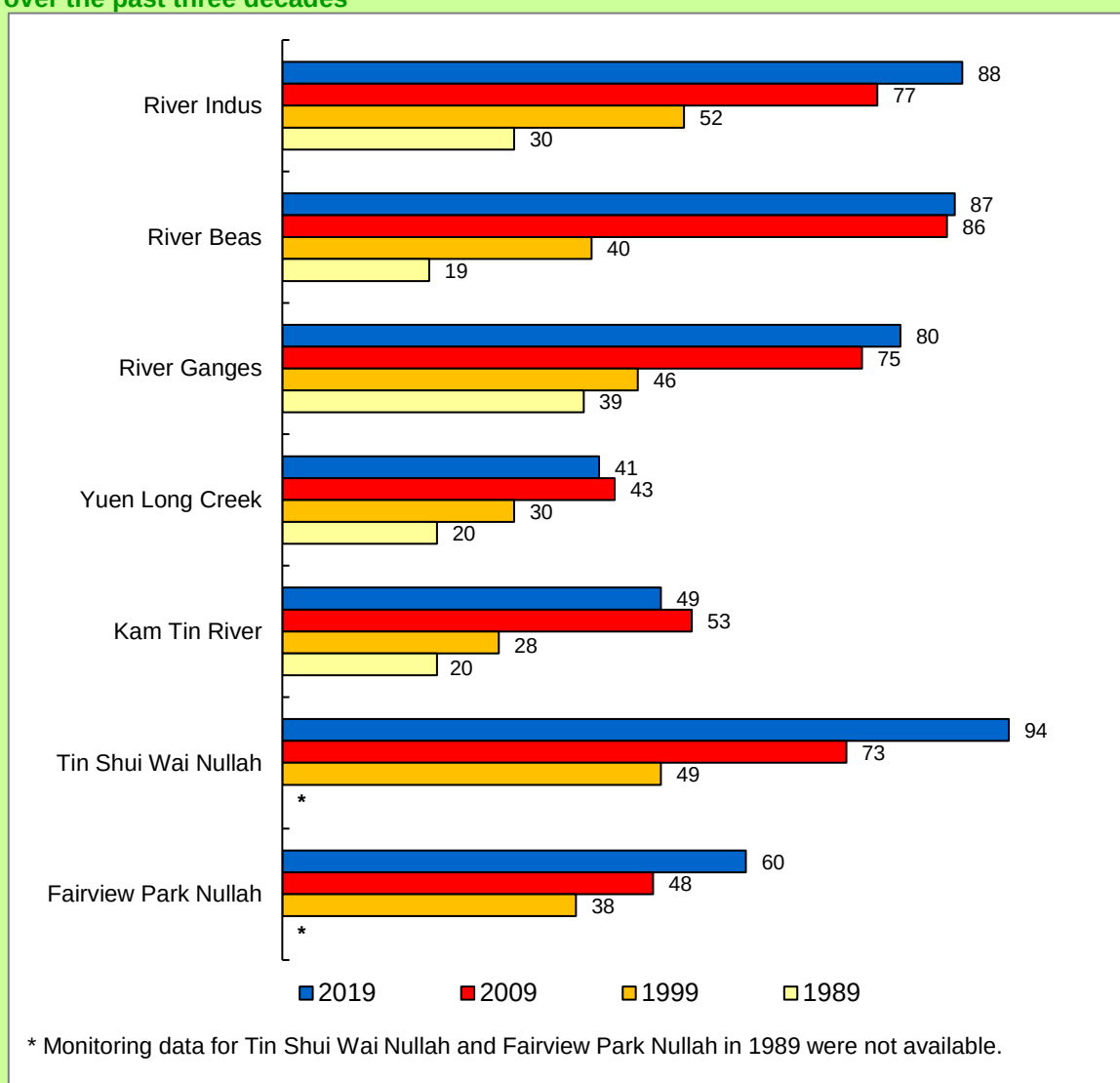
Water quality of the rivers and streams in the Northwestern New Territories has showed noticeable improvements over the past three decades. In 2019, the overall WQO compliance rate of all monitoring stations in these districts was 78%, as compared with 53% in 1999 and 39% in 1989 (Figure 14).

River Indus is a major river in the North District. It collects surface runoffs from densely populated Fanling and Sheung Shui urban areas before joining River Beas to drain into Shenzhen River. The overall WQO compliance rate was 88% in 2019, as compared with 30% in 1989. The downstream monitoring station (IN1) was upgraded from “Fair” in 2018 to “Good” in 2019. The mid-stream station (IN2) and upstream station (IN3) maintained to be “Good” and “Excellent” respectively in 2019 (Figure 16).

As a tributary of River Indus, River Beas recorded an overall WQO compliance rate of 87% in 2019, as compared with 19% in 1989. The upstream, mid-stream and downstream monitoring stations (RB1, RB2 and RB3) of the river achieved “Good” to “Excellent” gradings in 2019 (Figure 16), with improvement in both stations RB1 and RB3 as compared with 2018.

River Ganges had an overall WQO compliance rate of 80% in 2019, as compared with 39% in 1989. The upstream station (GR3) maintained “Excellent” in 2019, while both mid-stream station (GR2) and downstream station (GR1) were graded “Fair” (Figure 16).

Figure 14. WQO compliance rates (%) for rivers and streams in Northwestern New Territories over the past three decades



As a major river in Yuen Long District, Yuen Long Creek passes through rural areas as well as the densely populated Yuen Long new town and Yuen Long Kau Hui, before merging with Kam Tin River flowing into Deep Bay. In 2019, the overall WQO compliance rate for Yuen Long Creek was

41%, as compared with 20% in 1989. For Kam Tin River, the overall WQO compliance rate in 2019 was 49%, as compared with 20% in 1989.

The two upstream monitoring stations (YL1 and YL2) of Yuen Long Creek were graded as "Bad" and "Fair" respectively in 2019, as compared with both "Bad" in 2018. While the downstream stations (i.e., YL3 and YL4) were rated as "Bad" in 2019, as compared with "Very Bad" in 2018 (Figure 17). Both monitoring stations (KT1 and KT2) at Kam Tin River remained "Fair" and "Bad" respectively in 2019.

To improve the river water quality in the Northwestern New Territories, the Government is now carrying out a review of the local sewerage programmes with a view to further reducing the residual pollution load from unsewered villages. Measures include the provision of sewerage facilities for unsewered villages and installation of dry weather flow interceptors (DWFIs) where possible. The EPD has also continued to step up enforcement against illegal discharges and collaborating with the Agriculture, Fisheries and Conservation Department in undertaking education programmes to tackle the livestock farming related pollution in the area.



Fairview Park Nullah

Tin Shui Wai Nullah had an overall WQO compliance of 94% in 2019, as compared with 73% in 2009 and 49% in 1999. The upstream monitoring station (TSR2) and downstream station (TSR1) continued to be "Good" and "Fair" respectively in 2019 (Figure 17).

The Fairview Park Nullah (station FVR1) recorded a WQO compliance rate of 60% in 2019, as compared with 48% in 2009 and 38% in 1999. Its WQI grading remained “Fair” in 2019 (Figure 17).

In 2019, all 6 smaller streams in the Lau Fau Shan area maintained good water quality and 5 of them fully met the WQOs in 2019, as compared with 93% or above in 2018 (Figure 15). Sheung Pak Nai Stream (station DB5) also had a high WQO compliance rate of 98% in 2019. All these streams maintained “Excellent” in 2019 (Figure 18).

Figure 15. WQO compliance rates (%) for smaller streams in Lau Fau Shan area over the past three decades

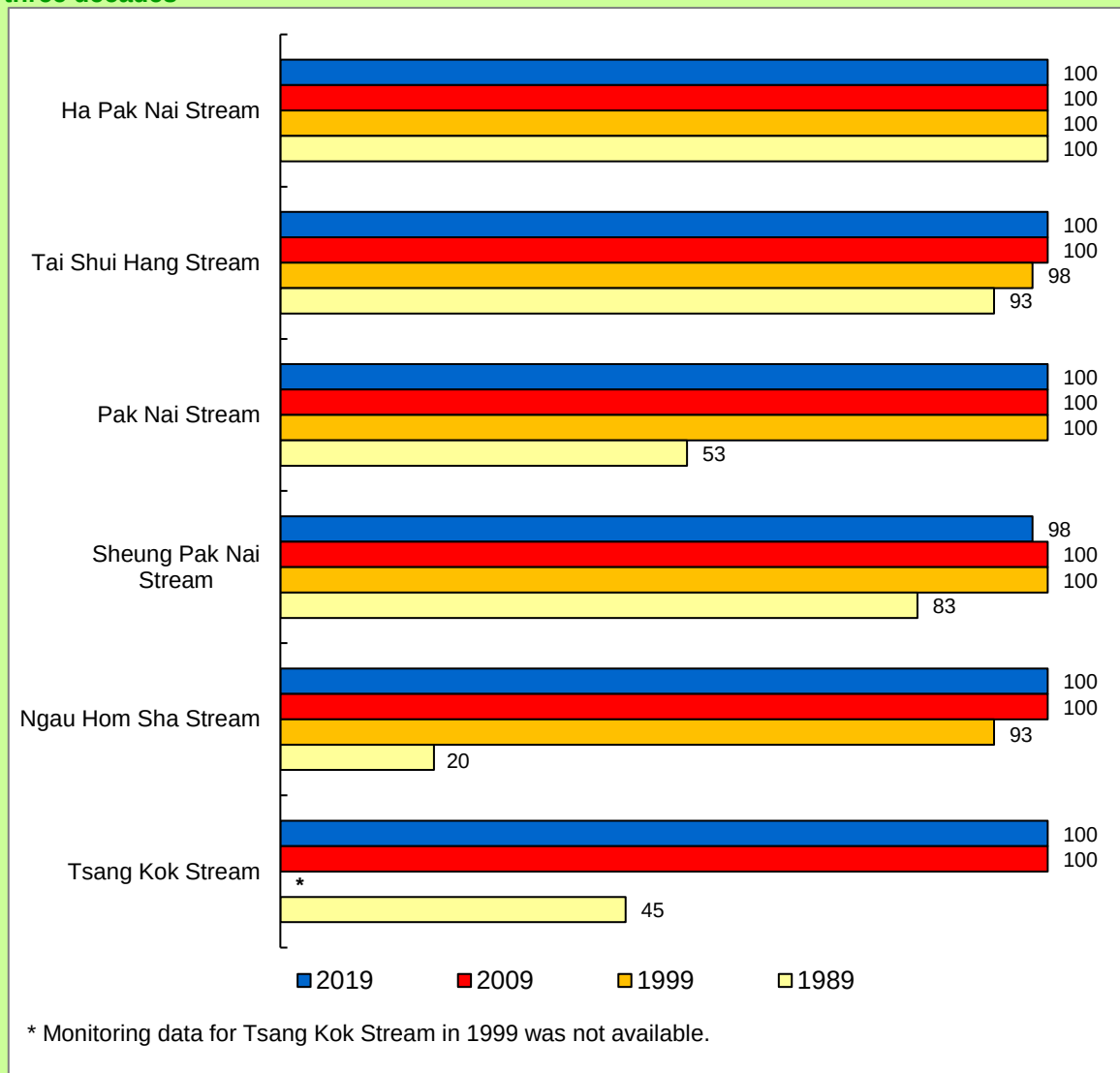


Figure 16. WQI gradings and *E. coli* levels in River Indus, River Beas and River Ganges

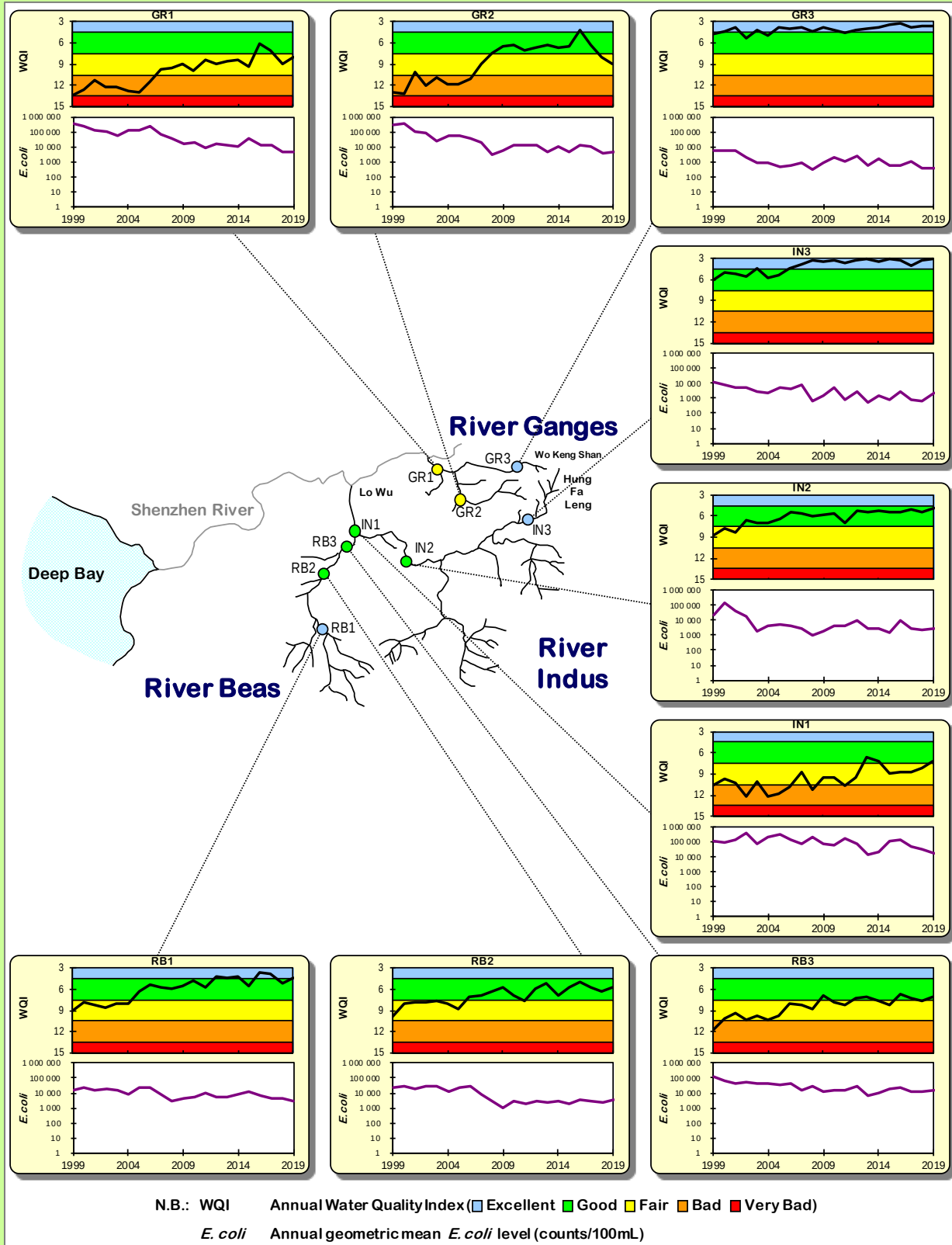


Figure 17. WQI gradings and *E. coli* levels in Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah

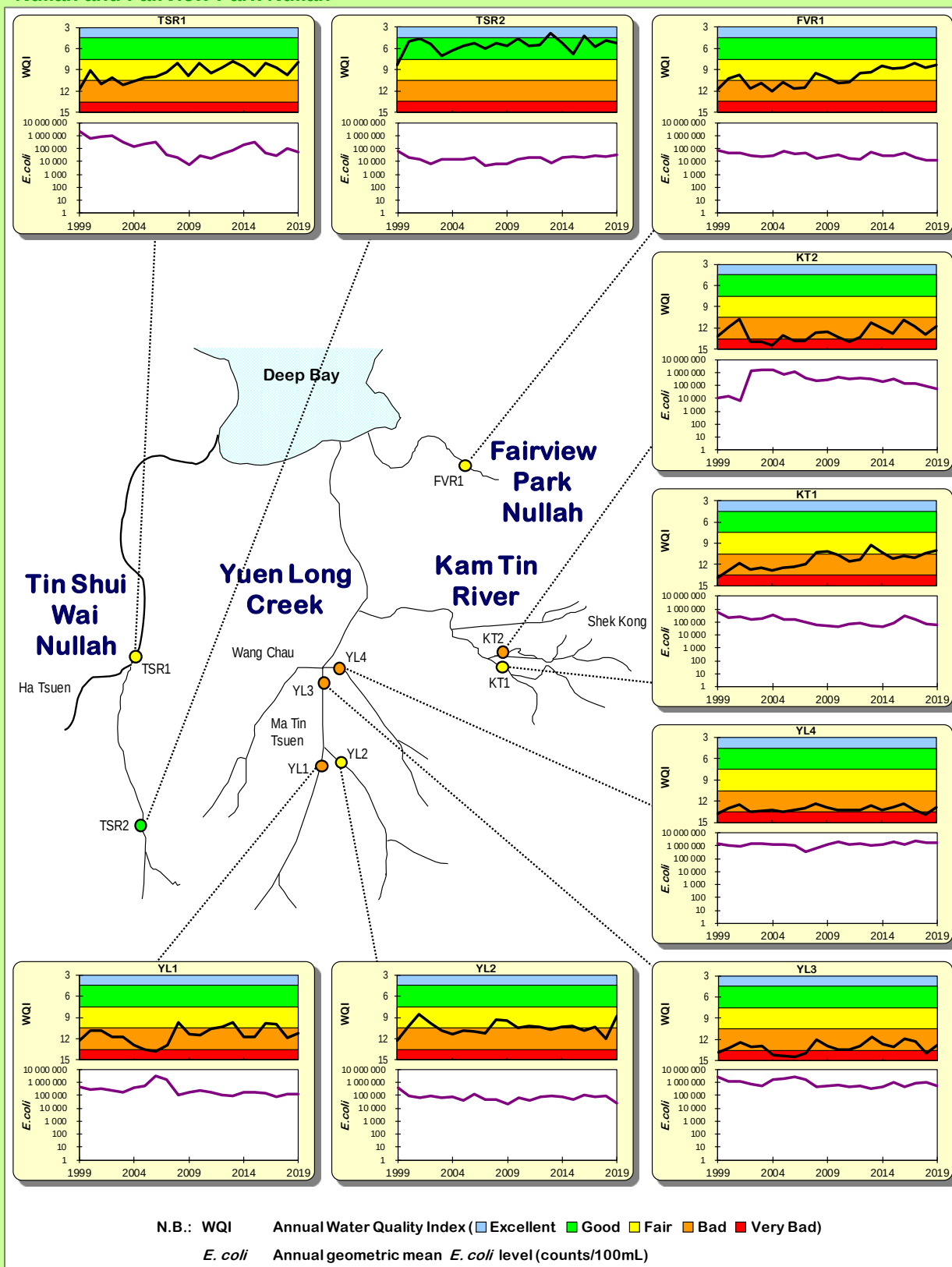
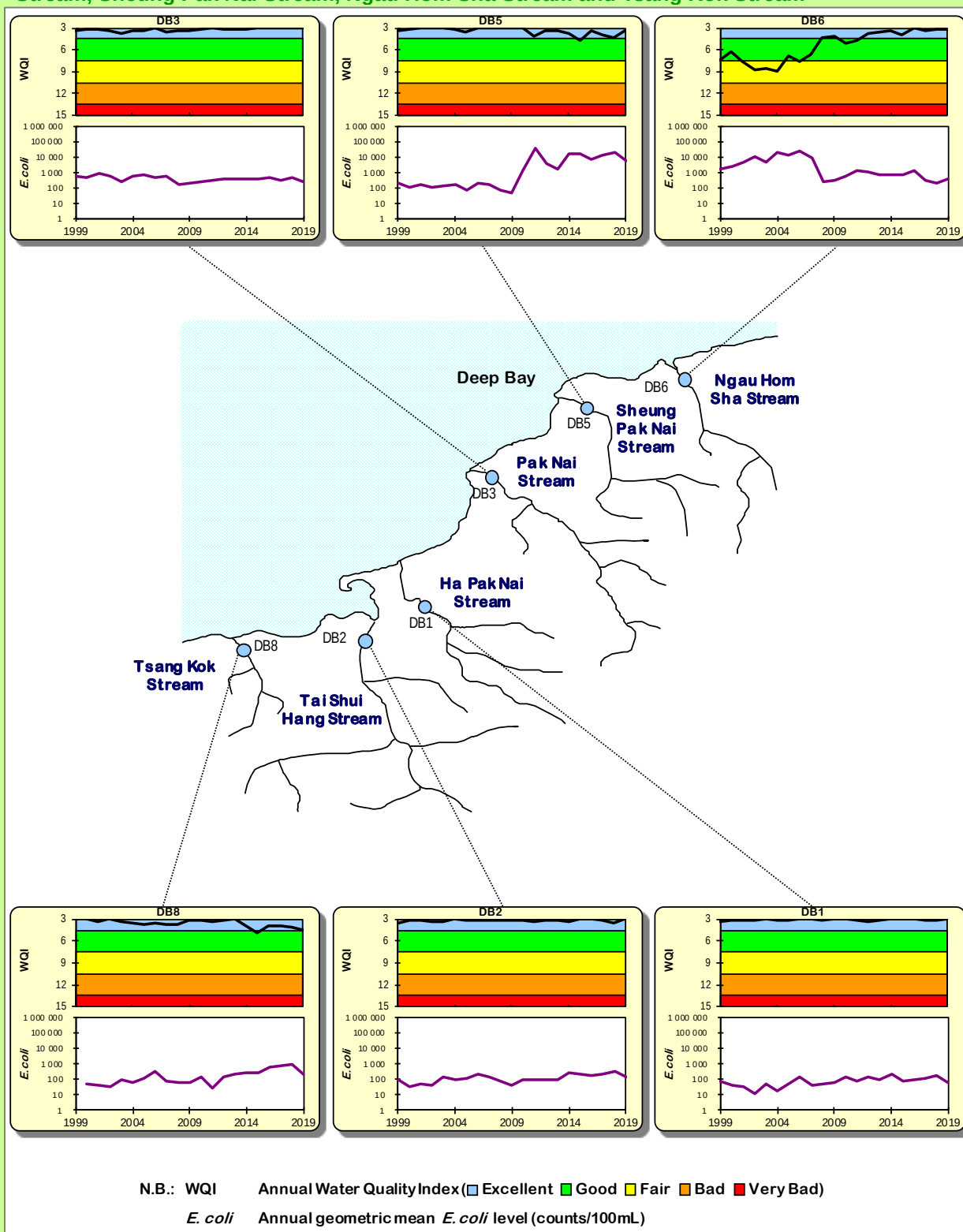


Figure 18. WQI gradings and *E. coli* levels in Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream and Tsang Kok Stream

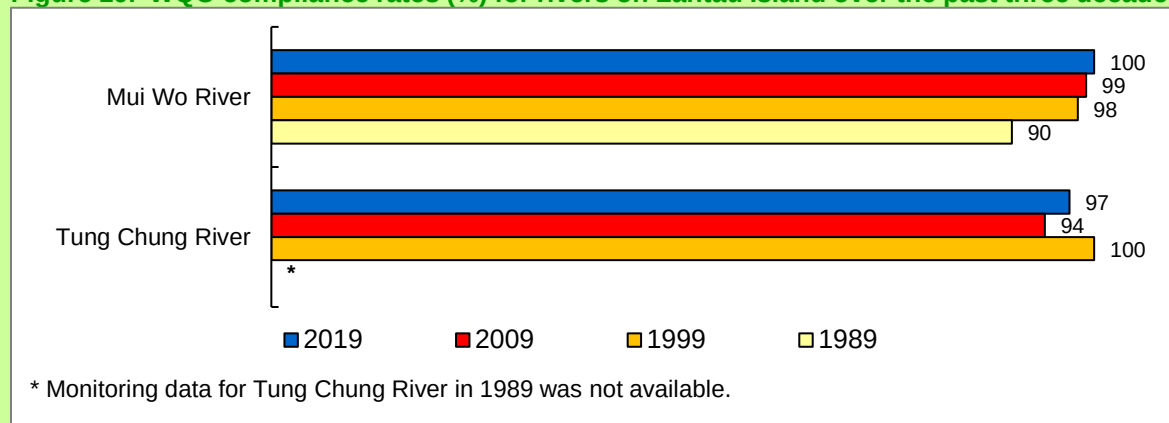


3.3. Lantau Island

Lantau Island has a relatively large land area but is less populated. Two rivers with a total of 8 monitoring stations are monitored by the EPD: 5 along Mui Wo River on the south-eastern side of Lantau Island (the Southern WCZ) and 3 at Tung Chung River on the north-western side of the island (the North Western WCZ).

The rivers in Lantau Island generally displayed satisfactory water quality over the past three decades. In 2019, the overall WQO compliance rate of all monitoring stations in the Lantau Island was 99%, as compared to 98% in 2009, 99% in 1999 and 80% in 1989. Mui Wo River achieved full WQO compliance (100%) in 2019, compared to 96% in 2018. The WQO compliance rate of Tung Chung River in 2019 was 97%, compared to 96% in 2018 (Figure 19). As for the WQI grading, while one monitoring station at Tung Chung River (TC3) was rated “Good” (Figures 20 and 21), all remaining stations at both rivers were graded “Excellent” in 2019. All stations maintained the same grading, except for the station MW5 at Mui Wo River displaying WQI upgrading in 2019.

Figure 19. WQO compliance rates (%) for rivers on Lantau Island over the past three decades





Mui Wo River

Figure 20. WQI gradings and *E. coli* levels in Mui Wo River

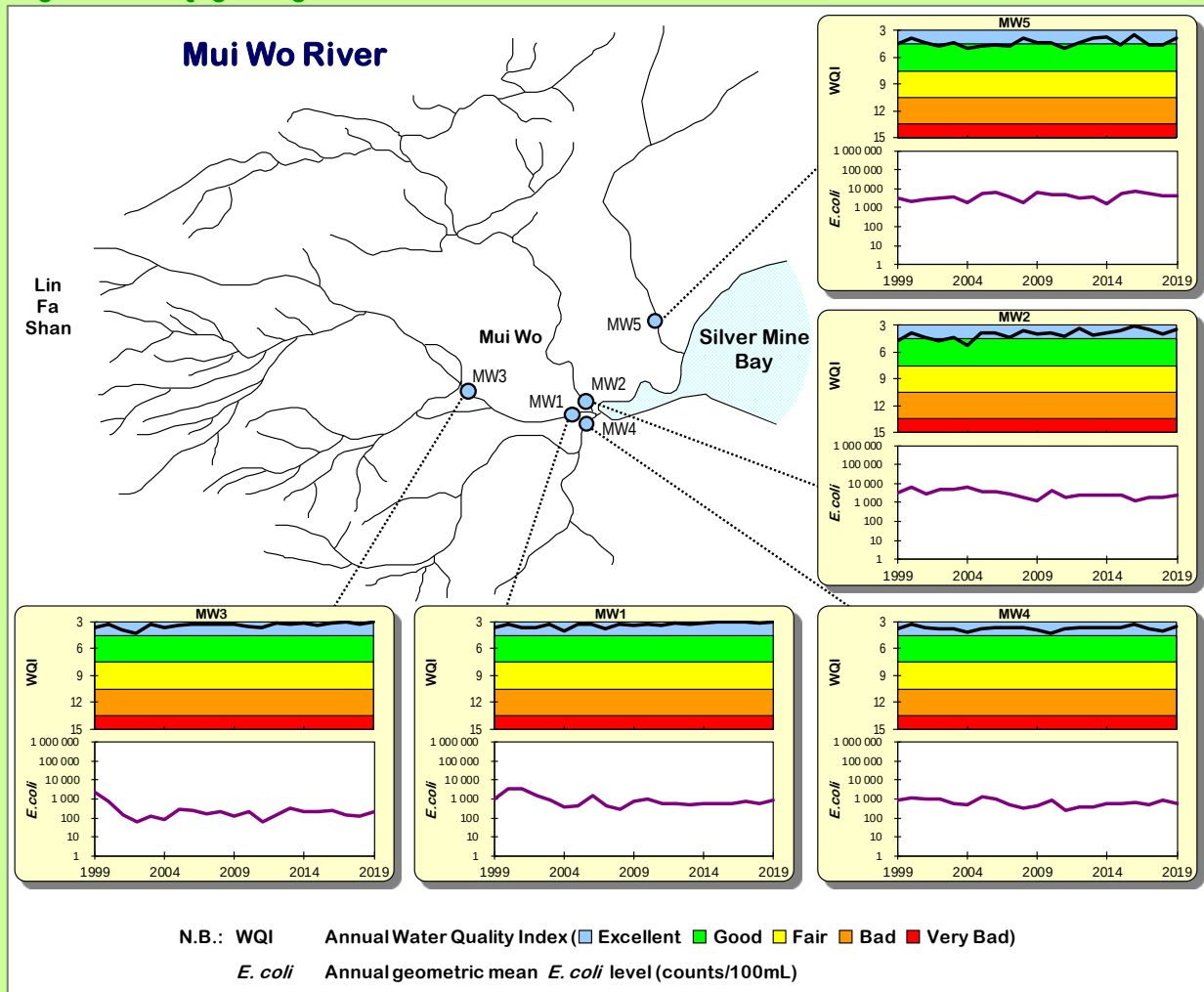
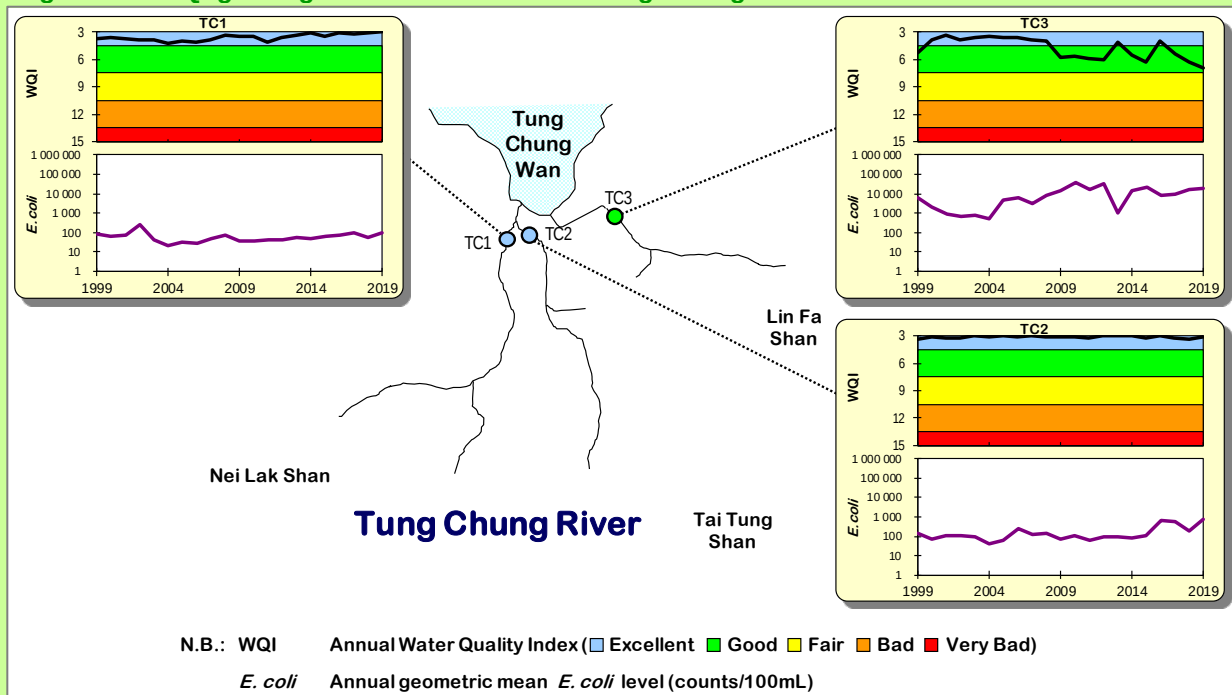


Figure 21. WQI gradings and *E. coli* levels in Tung Chung River



3.4. Southwestern New Territories and Kowloon

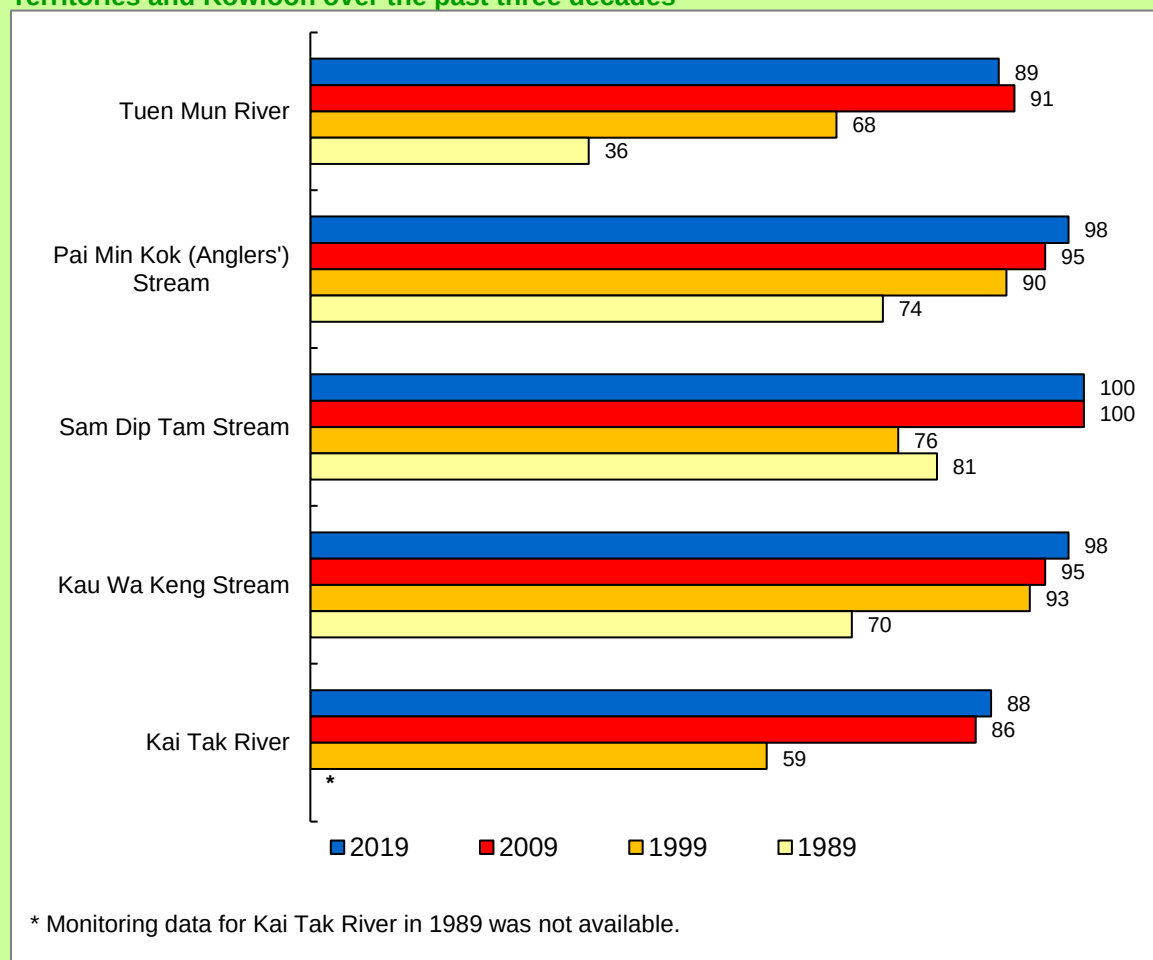
Five major watercourses in these areas being monitored by EPD include Tuen Mun River (in the North Western WCZ), Pai Min Kok Stream near Sham Tseng (in the Western Buffer WCZ), Sam Dip Tam Stream near Tsuen Wan, Kau Wa Keng Stream in Kwai Chung, and Kai Tak River located in East Kowloon (within the Victoria Harbour WCZ). There has been significant improvement in the water quality of these urban watercourses over the past 30 years (Figure 22).



Tuen Mun River

In 2019, 17 out of the 18 river monitoring stations (94%) in the Southwestern New Territories and Kowloon were graded “Excellent” or “Good” (Figures 23 to 25), same as 2018. Among these stations, only one station (i.e., the upstream station TN1 at Tuen Mun River) was graded “Bad” in 2019. The overall WQO compliance rate of the rivers and streams in the area has increased from 70% in 1999 to 92% in 2019.

Figure 22. WQO compliance rates (%) for rivers and streams in Southwestern New Territories and Kowloon over the past three decades



As a major river in the Southwestern New Territories, Tuen Mun River passes through Lam Tei, San Hing Tsuen and Fu Tei in its upstream section, and then the densely populated Tuen Mun town in its mid-stream before draining into the Tuen Mun Typhoon Shelter. The river showed marked water quality improvement in the last three decades, with its WQO compliance rate rising significantly from 36% in 1989 to 89% in 2019. Five of the 6 monitoring stations (TN2, TN3, TN4, TN5 and TN6) situated from the middle to lower sections of the river maintained “Good” WQI grading, attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun SMP. In 2019, the upstream monitoring station (TN1) was rated “Bad”, mainly due to discharges from unsewered rural areas (Figure 23). To further reduce pollution in Tuen Mun River, a DWFI has been installed since around 2001 near the West Rail Siu Hong Station to divert the contaminated river flow at TN1 to foul sewers leading to the sewage treatment works for treatment.

Pai Min Kok Stream near Sham Tseng achieved an overall WQO compliance rate of 98% in 2019, same as 2018. Both monitoring stations (AN1 and AN2) in the stream maintained “Excellent” grading in 2019 (Figure 24).

Sam Dip Tam Stream in Tsuen Wan fully met the WQOs in 2019. All 3 monitoring stations in the stream maintained “Excellent” grading in 2019 (Figure 24).

Kau Wa Keng Stream (monitoring station KW3) in Kwai Chung achieved a WQO compliance rate of 98% in 2019, as compared with 70% in 1989. Its “Excellent” grading continued in 2019 (Figure 24).

Kai Tak River in East Kowloon attained a WQO compliance rate of 88% in 2019, as compared with 59% in 1999. All 6 monitoring stations in the river maintained “Good” grading in 2019 (Figure 25). It is expected that the river water quality would further improve, upon completion of the sewerage repair and upgrading works in the river catchment area by the Drainage Service Department and other improvement works in the Kai Tak Development Area by the Civil Engineering and Development Department.



Kai Tak River

Figure 23. WQI gradings and *E. coli* levels in Tuen Mun River

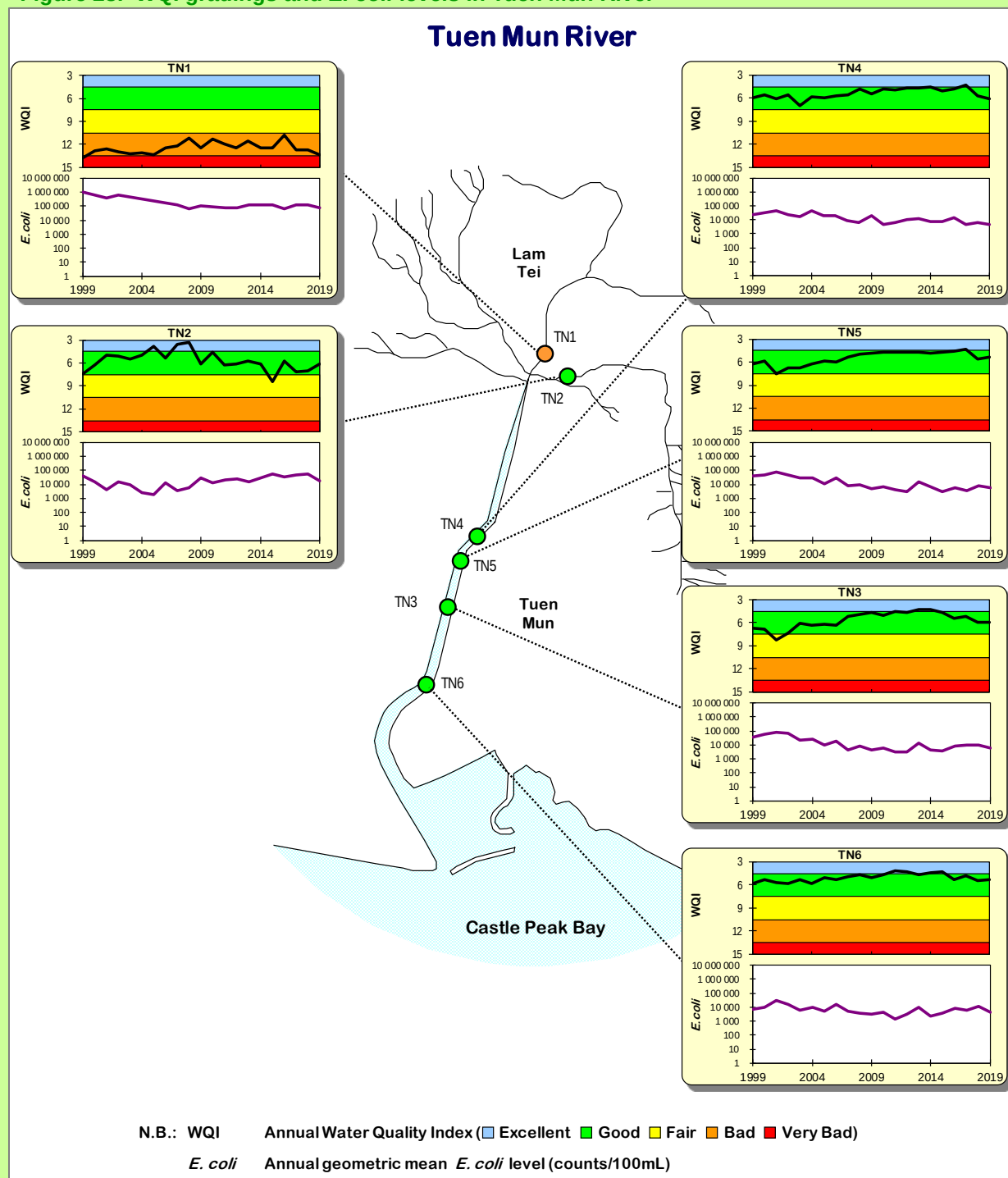


Figure 24. WQI gradings and *E. coli* levels in Pai Min Kok (Anglers') Stream, Sam Dip Tam Stream and Kau Wa Keng Stream

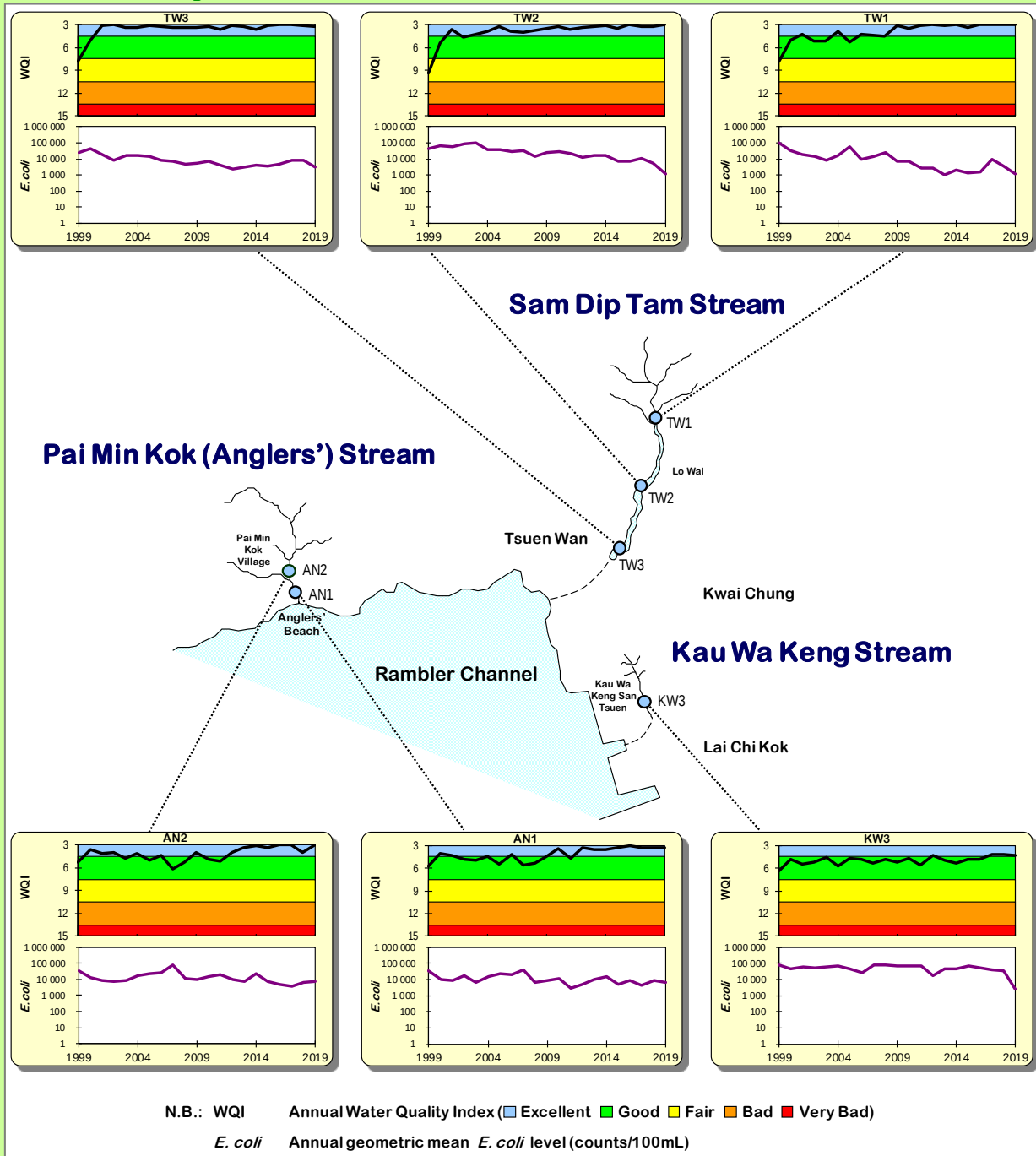
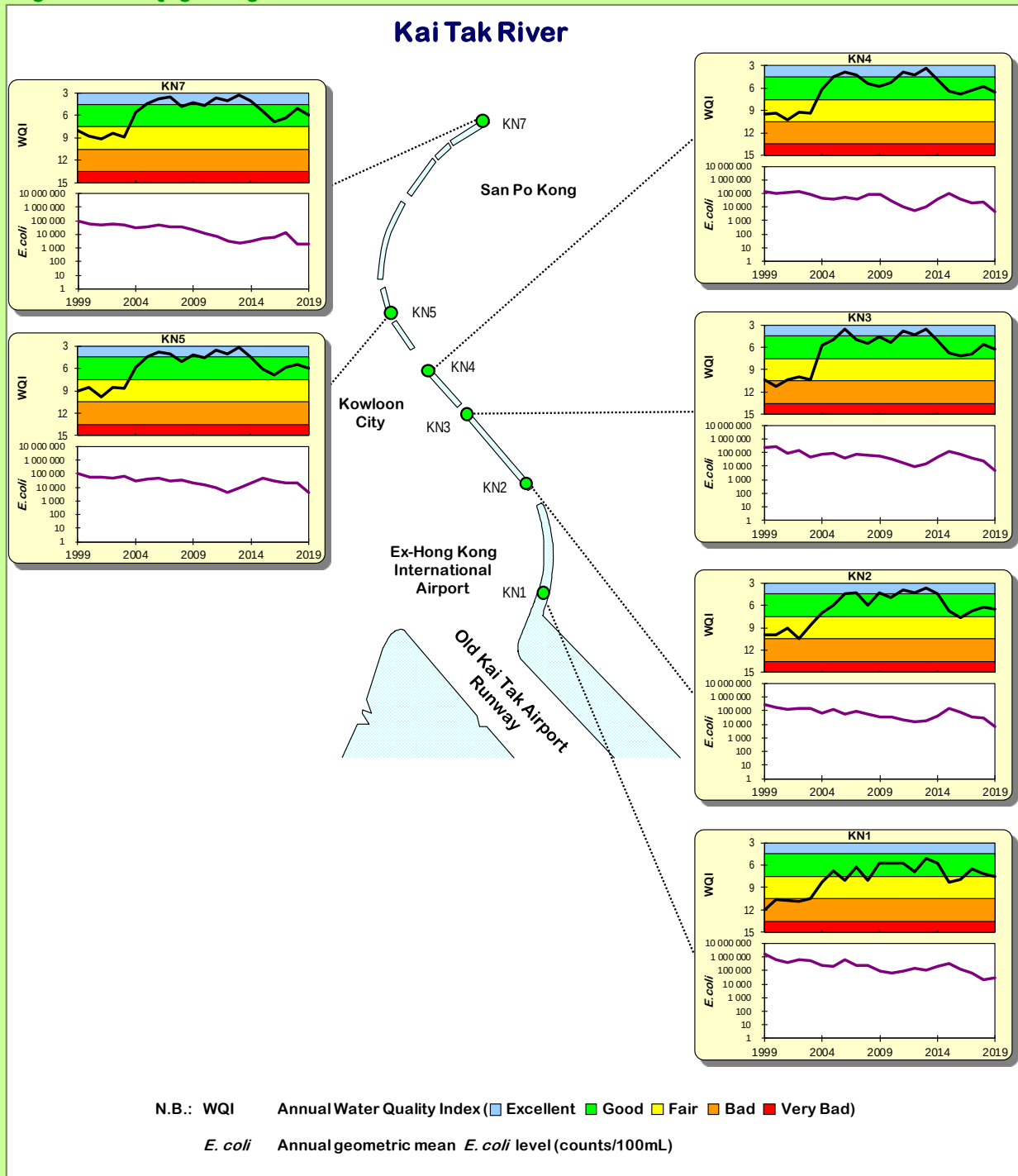


Figure 25. WQI gradings and *E. coli* levels in Kai Tak River



Appendices

Summary of river water quality monitoring stations and sampling frequency

Area	Watercourse	Monitoring Station (Number)	Sampling frequency
Eastern New Territories			
Sha Tin	Shing Mun River		
	<i>Shing Mun Main Channel</i>	TR19I (1)	monthly
	<i>Siu Lek Yuen Nullah</i>	TR23A, TR23L (2)	monthly
	<i>Fo Tan Nullah</i>	TR17, TR17L (2)	monthly
	<i>Kwun Yam Shan Stream</i>	KY1 (1)	monthly
	<i>Tai Wai Nullah</i>	TR19, TR19A, TR19C (3)	monthly
Tai Po Town Centre	<i>Tin Sum Nullah</i>	TR20B (1)	monthly
	Lam Tsuen River	TR12, TR12B, TR12C, TR12D, TR12E, TR12F, TR12G, TR12H, TR12I (9)	monthly
Tai Po Rural Area	Tai Po River	TR13 (1)	monthly
	Tai Po Kau Stream	TR14 (1)	monthly
	Shan Liu Stream	TR4 (1)	monthly
Sai Kung	Tung Tze Stream	TR6 (1)	monthly
	Ho Chung River	PR1, PR2 (2)	monthly
	Sha Kok Mei Stream	PR5, PR6 (2)	monthly
Tseung Kwan O	Tai Chung Hau Stream	PR7, PR8 (2)	monthly
	Tseng Lan Shue Stream	JR3, JR6, JR11 (3)	monthly
Northwestern New Territories			
North District	River Indus (Ng Tung River)	IN1, IN2, IN3 (3)	monthly
	River Beas (Sheung Yue River)	RB1, RB2, RB3 (3)	monthly
	River Ganges (Ping Yuen River)	GR1, GR2, GR3 (3)	monthly
Yuen Long	Yuen Long Creek	YL1, YL2, YL3, YL4 (4)	monthly
	Kam Tin River	KT1, KT2 (2)	monthly
	Tin Shui Wai Nullah	TSR1, TSR2 (2)	monthly
	Fairview Park Nullah	FVR1 (1)	monthly
Lau Fau Shan	Ha Pak Nai Stream	DB1 (1)	monthly
	Tai Shui Hang Stream	DB2 (1)	monthly
	Pak Nai Stream	DB3 (1)	monthly
	Sheung Pak Nai Stream	DB5 (1)	monthly
	Ngau Hom Sha Stream	DB6 (1)	monthly
	Tsang Kok Stream	DB8 (1)	monthly
Lantau Island			
Mui Wo	Mui Wo River	MW1, MW2, MW3, MW4, MW5 (5)	monthly
Tung Chung	Tung Chung River	TC1, TC2, TC3 (3)	monthly
Southwestern New Territories and Kowloon			
Tuen Mun	Tuen Mun River	TN1, TN2, TN3, TN4, TN5, TN6 (6)	monthly
Tsuen Wan and Kwai Chung	Pai Min Kok (Anglers') Stream	AN1, AN2 (2)	monthly
	Sam Dip Tam Stream	TW1, TW2, TW3 (3)	monthly
	Kau Wa Keng Stream	KW3 (1)	monthly
Kowloon	Kai Tak River	KN1, KN2, KN3, KN4, KN5, KN7 (6)	monthly
Total	30	82	-

River water quality parameters and analytical methods (part 1 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Physical Chemical Properties		
Water Temperature	0.1 °C	Multi-parameter water quality data logger, model YSI-6820 / model YSI ProDss / On-site measurement / EPD
Dissolved Oxygen	0.1 mg/L, 1% saturation	
pH	0.1	
Conductivity	1 µS/cm	
Salinity	0.01, ppt	
Turbidity	0.1 NTU	
Flow	0.001 m³/s	Global Water Flow Probe, model FP211 / Electromagnetic flow meter, model Hach FH950 / On-site measurement / EPD
Solid Contents		
Suspended Solids	0.5 mg/L	In-house method GL-PH-23, based on APHA 22ed 2540 D & E / Government Laboratory
Total Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 B & E / Government Laboratory
Total Volatile Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 B & E / Government Laboratory
Aggregate Organics		
5-Day Biochemical Oxygen Demand (BOD ₅)	0.1 mg/L	In-house method based on APHA 18ed 5210 B / EPD
Chemical Oxygen Demand (COD)	2 mg/L	In-house method GL-OR-38, based on ASTM D1252-00, Method A or in-house method GL-OR-39, based on ASTM D1252-00, Method B / Government Laboratory
Total Organic Carbon (TOC)	1 mg/L	In-house method GL-OR-32, based on APHA 21ed 5310 B / Government Laboratory
Faecal Bacteria		
<i>Escherichia coli</i> (<i>E. coli</i>)	1 count/100 mL	In-house method ² , membrane filtration with CHROMagar Liquid ECC medium / EPD
Faecal Coliforms	1 count/100 mL	
Nutrients		
Ammonia-Nitrogen	0.005 mg/L	In-house method GL-IN-15, based on ASTM Standards, D 3590-11 Test Method B / Government Laboratory
Nitrite-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₂ ⁻ B / Government Laboratory
Nitrate-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO ₃ ⁻ I / Government Laboratory
Total Kjeldahl Nitrogen	0.05 mg/L	In-house methods GL-IN-14 and GL-IN-15, based on ASTM D3590-11 Test Method B / Government Laboratory
Orthophosphate Phosphorus	0.002 mg/L	In-house method GL-IN-16, based on APHA 22ed 4500-P G / Government Laboratory
Total Phosphorus	0.02 mg/L	In-house methods GL-IN-14 and GL-IN-16, based on APHA 22ed 4500-P G, ASTM D515-88 B (FIA) / Government Laboratory
Silica (as SiO ₂)	0.05 mg/L	In-house method GL-IN-17, based on APHA 22ed 4500-SiO ₂ F / Government Laboratory

Reference notes:

1. Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.
2. i) Ho, B.S.W. and Tam, T.Y. (1997). Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. *Wat. Sci. Tech.*, **35**, 409-413.
ii) DoE and DHSS (1983). "The bacteriological examination of drinking water supplies 1982. Report on Public Health and Medical Subjects No. 71. Methods for the Examination of Waters and Associated Materials". Department of Environment, Department of Health and Social Security, Public Health Laboratory Service, H.M.S.O. London.

River water quality parameters and analytical methods (part 2 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method ¹ / Analyst
Metals		
Aluminium	50 µg/L	In-house methods GL-TE-63 and GL-TE-89, based on USEPA method 6020B (ICP-MS) / Government Laboratory
Antimony	1 µg/L	
Arsenic	1 µg/L	
Barium	1 µg/L	
Beryllium	1 µg/L	
Boron	50 µg/L	
Cadmium	0.1 µg/L	
Chromium	1 µg/L	
Copper	1 µg/L	
Iron	50 µg/L	
Lead	1 µg/L	
Manganese	10 µg/L	
Mercury	1 µg/L	
Molybdenum	2 µg/L	
Nickel	1 µg/L	
Silver	1 µg/L	
Thallium	1 µg/L	
Vanadium	2 µg/L	
Zinc	10 µg/L	
Pollutants from Industrial and Commercial Sources		
Cyanide	0.01 mg/L	In-house method GL-IN-42, based on ASTM D 4374-06 / Government Laboratory
Chloride	10 mg/L	In-house method GL-IN-43, based on APHA 20ed 4500-Cl ⁻ E & G / Government Laboratory
Fluoride	0.2 mg/L	In-house method GL-IN-47, based on APHA 20ed 4500-F ⁻ C & G / Government Laboratory
Anionic Surfactants (as Manoxol OT)	0.05 mg/L	In-house method GL-OR-30, based on BS 6068, Section 2.23 (1986), BS EN 903:1994, BS 6068: Section 2.23:1994 (Colorimetric) & In-house method GL-OR-65, based on Abbott, D.C. “Analyst”, Vol.87, p.286(1962) & S. Motomizu et al., “Analyst” Vol.113, p.747(1988) (FIA) / Government Laboratory
Oil and Grease	0.5 mg/L	In-house method GL-OR-26, based on APHA 20ed 5520 C / Government Laboratory
Sulphide Contents		
Free Hydrogen Sulphide	0.01 mg/L	In-house method GL-IN-46, based on APHA 20ed 4500S ²⁻ D / Government Laboratory
Sulphide	0.02 mg/L	
Plant Pigments		
Chlorophyll-a	0.2 µg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 / Government Laboratory
Phaeopigment	0.2 µg/L	

Key WQOs for river monitoring stations in Eastern New Territories

Watercourse	Monitoring station	Key Water Quality Objectives (WQOs)				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Tolo Harbour and Channel Water Control Zone						
Shing Mun River	KY1	6.5 - 8.5	3	15	20	4
	TR17	6.5 - 8.5	5	30	20	4
	TR17L	6.5 - 8.5	5	30	20	4
	TR19	6.5 - 8.5	5	30	20	4
	TR19A	6.5 - 8.5	5	30	20	4
	TR19C	6.5 - 8.5	5	30	20	4
	TR19I	6.0 - 9.0	5	30	25	4
	TR20B	6.5 - 8.5	5	30	20	4
	TR23A	6.5 - 8.5	3	15	20	4
	TR23L	6.5 - 8.5	3	15	20	4
Lam Tsuen River	TR12	6.5 - 8.5	3	15	20	4
	TR12B	6.5 - 8.5	3	15	20	4
	TR12C	6.5 - 8.5	3	15	20	4
	TR12D	6.5 - 8.5	3	15	20	4
	TR12E	6.5 - 8.5	3	15	20	4
	TR12F	6.5 - 8.5	3	15	20	4
	TR12G	6.5 - 8.5	3	15	20	4
	TR12H	6.5 - 8.5	3	15	20	4
	TR12I	6.0 - 9.0	5	30	25	4
Tai Po River	TR13	6.5 - 8.5	5	30	20	4
Tai Po Kau Stream	TR14	6.0 - 9.0	5	30	25	4
Shan Liu Stream	TR4	6.0 - 9.0	5	30	25	4
Tung Tze Stream	TR6	6.0 - 9.0	5	30	25	4
Port Shelter Water Control Zone						
Ho Chung River	PR1	6.5 - 8.5	5	30	25	4
	PR2	6.5 - 8.5	5	30	25	4
Sha Kok Mei Stream	PR5	6.0 - 9.0	5	30	25	4
	PR6	6.0 - 9.0	5	30	25	4
Tai Chung Hau Stream	PR7	6.0 - 9.0	5	30	25	4
	PR8	6.0 - 9.0	5	30	25	4
Junk Bay Water Control Zone						
Tseng Lan Shue Stream	JR3	6.0 - 9.0	5	30	25	4
	JR6	6.0 - 9.0	5	30	25	4
	JR11	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key WQOs for river monitoring stations in Northwestern New Territories

Watercourse	Monitoring station	Key Water Quality Objectives (WQOs)				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Deep Bay Water Control Zone						
River Indus	IN1	6.5 - 8.5	3	15	20	4
	IN2	6.5 - 8.5	3	15	20	4
	IN3	6.5 - 8.5	3	15	20	4
River Beas	RB1	6.5 - 8.5	3	15	20	4
	RB2	6.5 - 8.5	3	15	20	4
	RB3	6.5 - 8.5	3	15	20	4
River Ganges	GR1	6.5 - 8.5	3	15	20	4
	GR2	6.5 - 8.5	3	15	20	4
	GR3	6.5 - 8.5	3	15	20	4
Yuen Long Creek	YL1	6.5 - 8.5	3	15	20	4
	YL2	6.5 - 8.5	3	15	20	4
	YL3	6.5 - 8.5	5	30	20	4
	YL4	6.5 - 8.5	5	30	20	4
Kam Tin River	KT1	6.5 - 8.5	3	15	20	4
	KT2	6.5 - 8.5	3	15	20	4
Tin Shui Wai Nullah	TSR1	6.0 - 9.0	5	30	20	4
	TSR2	6.0 - 9.0	5	30	20	4
Fairview Park Nullah	FVR1	6.0 - 9.0	5	30	20	4
Ha Pak Nai Stream	DB1	6.0 - 9.0	5	30	20	4
Tai Shui Hang Stream	DB2	6.0 - 9.0	5	30	20	4
Pak Nai Stream	DB3	6.0 - 9.0	5	30	20	4
Sheung Pak Nai Stream	DB5	6.0 - 9.0	5	30	20	4
Ngau Hom Sha Stream	DB6	6.0 - 9.0	5	30	20	4
Tsang Kok Stream	DB8	6.0 - 9.0	5	30	20	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key WQOs for river monitoring stations on Lantau Island

Watercourse	Monitoring station	Key Water Quality Objectives (WQOs)				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids* (mg/L)	Minimum Dissolved Oxygen (mg/L)
Southern Water Control Zone						
Mui Wo River	MW1	6.5 - 8.5	5	30	20	4
	MW2	6.5 - 8.5	5	30	20	4
	MW3	6.5 - 8.5	5	30	20	4
	MW4	6.5 - 8.5	5	30	20	4
	MW5	6.0 - 9.0	5	30	25	4
North Western Water Control Zone						
Tung Chung River	TC1	6.0 - 9.0	5	30	25	4
	TC2	6.0 - 9.0	5	30	25	4
	TC3	6.0 - 9.0	5	30	25	4

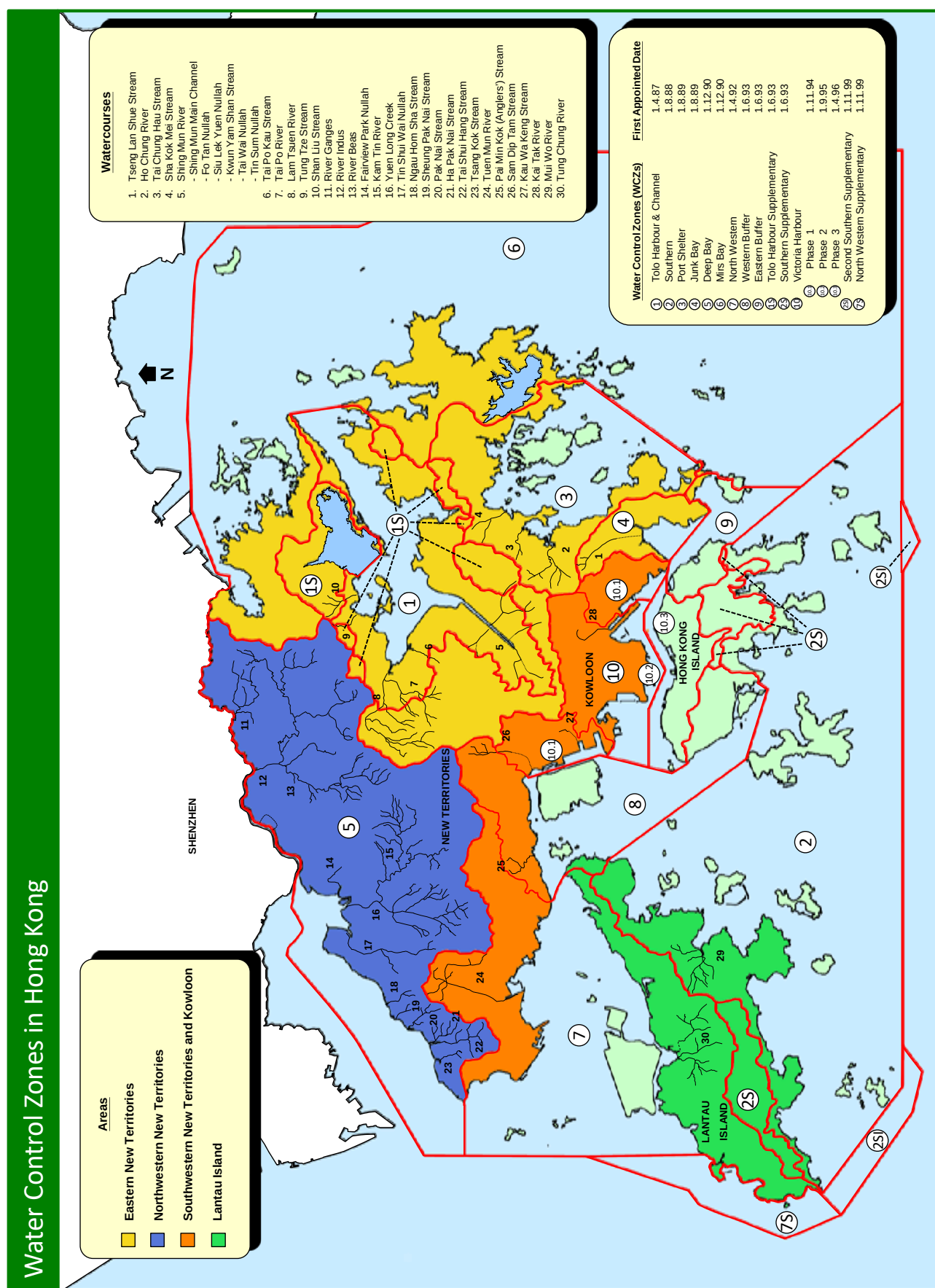
* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key WQOs for river monitoring stations in Southwestern New Territories and Kowloon

Watercourse	Monitoring station	Key Water Quality Objectives (WQOs)				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
North Western Water Control Zone						
Tuen Mun River	TN1	6.0 - 9.0	5	30	25	4
	TN2	6.5 - 8.5	3	15	20	4
	TN3	6.0 - 9.0	5	30	25	4
	TN4	6.0 - 9.0	5	30	25	4
	TN5	6.0 - 9.0	5	30	25	4
	TN6	6.0 - 9.0	5	30	25	4
Western Buffer Water Control Zone						
Pai Min Kok (Anglers') Stream	AN1	6.0 - 9.0	5	30	25	4
	AN2	6.0 - 9.0	5	30	25	4
Victoria Harbour Water Control Zone						
Sam Dip Tam Stream	TW1	6.0 - 9.0	5	30	25	4
	TW2	6.0 - 9.0	5	30	25	4
	TW3	6.0 - 9.0	5	30	25	4
Kau Wa Keng Stream	KW3	6.0 - 9.0	5	30	25	4
Kai Tak River	KN1	6.0 - 9.0	5	30	25	4
	KN2	6.0 - 9.0	5	30	25	4
	KN3	6.0 - 9.0	5	30	25	4
	KN4	6.0 - 9.0	5	30	25	4
	KN5	6.0 - 9.0	5	30	25	4
	KN7	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Water Control Zones (WCZs) in Hong Kong



Summary of water quality monitoring data for Shing Mun River (Main Channel and Siu Lek Yuen Nullah) in 2019

Parameter	Unit	Shing Mun Main Channel	Siu Lek Yuen Nullah	
		TR19I	TR23L	TR23A
Dissolved Oxygen	mg/L	7.7 (6.5 - 9.4)	8.3 (7.6 - 9.9)	7.3 (4.6 - 8.5)
pH		8.1 (7.7 - 8.4)	8.9 (8.6 - 9.0)	8.0 (7.7 - 8.3)
Suspended Solids	mg/L	8.5 (2.8 - 21.0)	1.6 (0.6 - 10.0)	6.5 (1.8 - 31.0)
5-Day Biochemical Oxygen Demand	mg/L	3.6 (1.4 - 7.1)	0.4 (<0.1 - 1.9)	1.7 (0.7 - 4.4)
Chemical Oxygen Demand	mg/L	13 (9 - 18)	3 (<2 - 13)	9 (5 - 22)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.3)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	520 (10 - 44 000)	200 (40 - 1 800)	6 400 (1 100 - 49 000)
Faecal Coliforms	counts/ 100mL	4 500 (410 - 48 000)	3 000 (270 - 69 000)	18 000 (4 400 - 100 000)
Ammonia-Nitrogen	mg/L	0.125 (0.041 - 0.290)	0.014 (<0.005 - 0.050)	0.150 (0.089 - 0.580)
Nitrate-Nitrogen	mg/L	0.140 (<0.002 - 0.510)	0.240 (0.190 - 0.840)	0.365 (0.150 - 0.550)
Total Kjeldahl Nitrogen	mg/L	0.49 (0.28 - 0.92)	0.15 (<0.05 - 0.55)	0.47 (0.23 - 1.10)
Orthophosphate Phosphorus	mg/L	0.018 (<0.002 - 0.030)	0.004 (<0.002 - 0.100)	0.021 (<0.002 - 0.056)
Total Phosphorus	mg/L	0.05 (0.03 - 0.07)	<0.02 (<0.02 - 0.12)	0.04 (0.03 - 0.09)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	<50 (<50 - 121)	<50 (<50 - 148)	<50 (<50 - 542)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 4)	<1 (<1 - <1)	2 (<1 - 2)
Copper	µg/L	4 (3 - 6)	<1 (<1 - 4)	3 (2 - 31)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	14 (<10 - 41)	<10 (<10 - 30)	16 (<10 - 39)
Flow	m ³ /s	NM	0.031 (0.009 - 0.080)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Fo Tan Nullah and Kwun Yam Shan Stream) in 2019

Parameter	Unit	<u>Fo Tan Nullah</u>		<u>Kwun Yam Shan Stream</u>
		TR17	TR17L	KY1
Dissolved Oxygen	mg/L	9.1 (7.5 - 10.2)	6.8 (4.7 - 8.3)	8.5 (7.6 - 9.6)
pH		9.3 (7.5 - 10.0)	7.7 (7.4 - 8.0)	8.3 (7.7 - 8.6)
Suspended Solids	mg/L	3.0 (1.4 - 15.0)	5.8 (3.2 - 34.0)	7.5 (2.1 - 21.0)
5-Day Biochemical Oxygen Demand	mg/L	1.7 (0.4 - 17.0)	1.7 (0.9 - 5.2)	0.3 (<0.1 - 0.6)
Chemical Oxygen Demand	mg/L	8 (3 - 14)	12 (5 - 18)	6 (<2 - 10)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	2 000 (700 - 19 000)	2 900 (60 - 200 000)	380 (57 - 5 800)
Faecal Coliforms	counts/ 100mL	14 000 (3 000 - 90 000)	27 000 (840 - 3 100 000)	1 500 (590 - 8 500)
Ammonia-Nitrogen	mg/L	0.043 (0.016 - 0.660)	0.215 (0.180 - 0.870)	0.021 (0.012 - 0.047)
Nitrate-Nitrogen	mg/L	0.475 (0.180 - 1.000)	0.345 (0.150 - 0.630)	0.530 (0.310 - 0.930)
Total Kjeldahl Nitrogen	mg/L	0.40 (0.11 - 3.00)	0.59 (0.31 - 1.40)	0.33 (0.05 - 1.80)
Orthophosphate Phosphorus	mg/L	0.021 (0.004 - 0.059)	0.026 (0.007 - 0.084)	0.065 (0.013 - 0.099)
Total Phosphorus	mg/L	0.05 (0.03 - 0.19)	0.06 (<0.02 - 0.16)	0.10 (0.07 - 0.18)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	154 (<50 - 260)	<50 (<50 - 161)	162 (70 - 590)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	2 (1 - 3)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 28)	4 (2 - 4)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 17)	16 (<10 - 33)	<10 (<10 - 16)
Flow	m ³ /s	0.075 (0.020 - 0.425)	NM	0.008 (0.000 - 0.086)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Tai Wai Nullah and Tin Sum Nullah) in 2019

Parameter	Unit	Tai Wai Nullah		Tin Sum Nullah	
		TR19A	TR19C	TR19	TR20B
Dissolved Oxygen	mg/L	9.0 (8.3 - 10.8)	9.4 (8.4 - 11.1)	10.5 (8.7 - 13.0)	8.2 (7.8 - 9.2)
pH		7.8 (7.2 - 8.4)	7.5 (7.2 - 7.8)	7.5 (7.1 - 8.3)	7.3 (6.9 - 9.7)
Suspended Solids	mg/L	3.0 (1.8 - 7.0)	1.9 (1.0 - 13.0)	2.8 (1.1 - 7.7)	3.2 (0.6 - 390.0)
5-Day Biochemical Oxygen Demand	mg/L	0.8 (0.3 - 2.9)	1.0 (0.6 - 6.8)	2.1 (0.9 - 5.9)	<0.1 (<0.1 - 1.0)
Chemical Oxygen Demand	mg/L	7 (3 - 13)	6 (3 - 18)	8 (4 - 12)	5 (2 - 26)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	320 (<1 - 16 000)	1 400 (180 - 14 000)	3 400 (150 - 81 000)	1 (<1 - 6)
Faecal Coliforms	counts/ 100mL	1 400 (<1 - 51 000)	14 000 (4 400 - 84 000)	82 000 (13 000 - 280 000)	2 (<1 - 22)
Ammonia-Nitrogen	mg/L	0.042 (<0.005 - 0.140)	0.042 (0.012 - 0.530)	0.037 (0.010 - 0.530)	0.069 (<0.005 - 0.380)
Nitrate-Nitrogen	mg/L	1.000 (0.810 - 1.600)	0.665 (0.470 - 1.600)	0.635 (0.520 - 1.100)	1.010 (0.410 - 1.800)
Total Kjeldahl Nitrogen	mg/L	0.43 (0.21 - 0.86)	0.40 (0.18 - 1.20)	0.45 (0.31 - 0.88)	0.39 (0.13 - 1.90)
Orthophosphate Phosphorus	mg/L	0.013 (0.003 - 0.028)	0.016 (0.004 - 0.170)	0.007 (<0.002 - 0.130)	0.010 (0.003 - 0.110)
Total Phosphorus	mg/L	0.04 (<0.02 - 0.05)	0.03 (0.02 - 0.22)	0.03 (0.02 - 0.16)	<0.02 (<0.02 - 0.81)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.06)
Aluminium	µg/L	98 (<50 - 443)	69 (<50 - 210)	61 (<50 - 122)	55 (<50 - 1 612)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.5)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 2)	<1 (<1 - 2)
Copper	µg/L	<1 (<1 - 4)	3 (<1 - 7)	3 (<1 - 4)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	22 (<10 - 52)	19 (<10 - 39)	17 (<10 - 31)	11 (<10 - 16)
Flow	m ³ /s	0.025 (0.014 - 0.095)	0.060 (0.024 - 0.180)	0.150 (0.045 - 0.384)	0.026 (0.012 - 0.054)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2019 (Part 1 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12H	TR12D	TR12C
Dissolved Oxygen	mg/L	8.6 (7.7 - 9.9)	8.7 (7.8 - 10.1)	8.3 (7.9 - 9.6)
pH		7.6 (7.2 - 7.7)	7.6 (7.0 - 7.7)	7.6 (7.2 - 7.8)
Suspended Solids	mg/L	1.8 (<0.5 - 3.2)	1.1 (0.7 - 2.5)	2.7 (1.2 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	0.4 (<0.1 - 1.7)	0.3 (<0.1 - 1.7)	1.3 (0.8 - 3.6)
Chemical Oxygen Demand	mg/L	3 (<2 - 6)	<2 (<2 - 5)	6 (3 - 11)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 400 (140 - 10 000)	260 (62 - 4 100)	5 500 (1 700 - 60 000)
Faecal Coliforms	counts/ 100mL	4 300 (410 - 30 000)	3 000 (810 - 25 000)	19 000 (6 800 - 120 000)
Ammonia-Nitrogen	mg/L	0.032 (0.018 - 0.055)	0.016 (0.010 - 0.039)	0.140 (0.048 - 0.410)
Nitrate-Nitrogen	mg/L	0.610 (0.380 - 0.920)	0.295 (0.086 - 0.460)	0.785 (0.430 - 1.100)
Total Kjeldahl Nitrogen	mg/L	0.26 (0.14 - 0.65)	0.19 (<0.05 - 0.60)	0.46 (0.30 - 1.00)
Orthophosphate Phosphorus	mg/L	0.029 (0.003 - 0.040)	0.011 (<0.002 - 0.019)	0.054 (0.015 - 0.084)
Total Phosphorus	mg/L	0.04 (0.02 - 0.05)	<0.02 (<0.02 - 0.04)	0.09 (0.06 - 0.12)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 171)	<50 (<50 - 56)	<50 (<50 - 687)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 4)	<1 (<1 - 1)	<1 (<1 - 5)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	10 (<10 - 27)	16 (<10 - 44)	<10 (<10 - 56)
Flow	m³/s	0.287 (0.088 - 1.140)	0.177 (0.030 - 0.435)	0.093 (0.040 - 0.234)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2019 (Part 2 of 3)

Parameter	Unit	<u>Lam Tsuen River</u>		
		TR12G	TR12F	TR12B
Dissolved Oxygen	mg/L	7.9 (7.2 - 9.6)	8.5 (7.9 - 9.7)	9.1 (8.3 - 10.3)
pH		7.4 (6.9 - 7.6)	7.6 (7.3 - 7.8)	7.5 (7.2 - 7.9)
Suspended Solids	mg/L	2.5 (0.9 - 5.8)	1.6 (<0.5 - 3.1)	1.6 (<0.5 - 2.5)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 1.6)	0.5 (0.1 - 1.8)	0.5 (0.1 - 1.4)
Chemical Oxygen Demand	mg/L	4 (<2 - 11)	5 (2 - 10)	4 (<2 - 9)
Oil & Grease	mg/L	<0.5 (<0.5 - 3.2)	<0.5 (<0.5 - 1.0)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	310 (72 - 5 700)	510 (59 - 19 000)	590 (230 - 3 900)
Faecal Coliforms	counts/ 100mL	2 100 (510 - 29 000)	4 500 (930 - 37 000)	4 900 (610 - 33 000)
Ammonia-Nitrogen	mg/L	0.016 (0.010 - 0.030)	0.027 (0.017 - 0.051)	0.042 (0.025 - 0.062)
Nitrate-Nitrogen	mg/L	0.062 (<0.002 - 0.160)	0.300 (0.200 - 0.650)	0.560 (0.220 - 1.000)
Total Kjeldahl Nitrogen	mg/L	0.17 (0.11 - 0.48)	0.25 (0.16 - 0.55)	0.27 (0.15 - 0.47)
Orthophosphate Phosphorus	mg/L	0.015 (0.002 - 0.024)	0.021 (0.007 - 0.091)	0.033 (0.003 - 0.050)
Total Phosphorus	mg/L	0.03 (0.02 - 0.05)	0.04 (0.02 - 0.13)	0.04 (0.02 - 0.05)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 102)	<50 (<50 - 63)	<50 (<50 - 94)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	<10 (<10 - 19)	<10 (<10 - 18)	<10 (<10 - 46)
Flow	m ³ /s	0.078 (0.025 - 0.164)	0.048 (0.009 - 0.224)	0.371 (0.108 - 1.944)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River (Part 3 of 3) and Tai Po River in 2019

Parameter	Unit	Lam Tsuen River			Tai Po River
		TR12E	TR12	TR12I	TR13
Dissolved Oxygen	mg/L	8.5 (7.7 - 9.4)	8.6 (6.1 - 9.5)	6.0 (3.6 - 6.6)	8.5 (8.0 - 9.9)
pH		8.1 (7.9 - 8.6)	7.6 (7.4 - 8.3)	7.2 (7.1 - 7.4)	7.5 (7.3 - 7.8)
Suspended Solids	mg/L	2.6 (0.5 - 4.9)	5.6 (1.1 - 18.0)	4.7 (1.6 - 7.1)	3.2 (0.6 - 5.5)
5-Day Biochemical Oxygen Demand	mg/L	0.4 (<0.1 - 1.7)	4.0 (0.7 - 10.0)	1.9 (0.7 - 3.2)	0.9 (0.5 - 3.0)
Chemical Oxygen Demand	mg/L	4 (<2 - 6)	12 (4 - 27)	9 (5 - 25)	5 (3 - 9)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.1)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.4)
<i>E. coli</i>	counts/ 100mL	1 300 (51 - 15 000)	2 600 (300 - 12 000)	45 000 (10 000 - 340 000)	16 000 (2 400 - 86 000)
Faecal Coliforms	counts/ 100mL	6 300 (2 400 - 46 000)	9 300 (1 800 - 29 000)	160 000 (65 000 - 620 000)	35 000 (5 700 - 360 000)
Ammonia-Nitrogen	mg/L	0.057 (0.018 - 0.099)	0.685 (0.180 - 4.300)	0.460 (0.260 - 0.890)	0.099 (0.030 - 0.240)
Nitrate-Nitrogen	mg/L	0.650 (0.420 - 1.100)	1.650 (0.560 - 3.500)	0.560 (0.350 - 0.980)	0.720 (0.460 - 1.100)
Total Kjeldahl Nitrogen	mg/L	0.30 (0.11 - 0.61)	1.60 (0.57 - 5.40)	0.94 (0.45 - 1.40)	0.54 (0.22 - 0.73)
Orthophosphate Phosphorus	mg/L	0.020 (<0.002 - 0.093)	0.255 (0.084 - 0.770)	0.078 (0.018 - 0.100)	0.051 (0.026 - 0.067)
Total Phosphorus	mg/L	0.03 (<0.02 - 0.12)	0.38 (0.10 - 0.91)	0.12 (0.07 - 0.15)	0.07 (0.04 - 0.09)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	90 (<50 - 783)	<50 (<50 - 122)	<50 (<50 - 113)	<50 (<50 - 156)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	2 (<1 - 2)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 3)	<1 (<1 - 2)	2 (2 - 4)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 25)	<10 (<10 - 29)	16 (<10 - 24)	10 (<10 - 20)
Flow	m ³ /s	0.307 (0.045 - 0.501)	0.124 (0.072 - 1.553)	NM	0.384 (0.104 - 0.738)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream in 2019

Parameter	Unit	<u>Tai Po Kau Stream</u>	<u>Shan Liu Stream</u>	<u>Tung Tze Stream</u>
		TR14	TR4	TR6
Dissolved Oxygen	mg/L	7.9 (6.3 - 9.7)	8.8 (8.0 - 9.9)	6.2 (3.0 - 8.0)
pH		7.0 (6.7 - 7.6)	8.0 (7.4 - 8.6)	7.4 (6.9 - 7.9)
Suspended Solids	mg/L	1.6 (0.9 - 5.7)	3.7 (1.0 - 5.7)	7.8 (2.8 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	0.2 (<0.1 - 0.6)	0.6 (0.3 - 1.4)	0.9 (0.5 - 1.5)
Chemical Oxygen Demand	mg/L	4 (2 - 42)	3 (<2 - 7)	11 (4 - 24)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.9)
<i>E. coli</i>	counts/ 100mL	640 (120 - 7 500)	1 300 (240 - 15 000)	2 400 (200 - 29 000)
Faecal Coliforms	counts/ 100mL	1 900 (250 - 13 000)	6 300 (3 200 - 28 000)	4 100 (220 - 33 000)
Ammonia-Nitrogen	mg/L	0.080 (0.037 - 0.150)	0.060 (0.013 - 0.190)	0.410 (0.160 - 1.200)
Nitrate-Nitrogen	mg/L	0.275 (0.130 - 0.560)	0.485 (0.220 - 0.970)	0.325 (0.076 - 0.850)
Total Kjeldahl Nitrogen	mg/L	0.27 (0.08 - 0.64)	0.26 (0.16 - 0.63)	0.72 (0.42 - 1.30)
Orthophosphate Phosphorus	mg/L	0.011 (<0.002 - 0.022)	0.035 (0.007 - 0.051)	0.041 (<0.002 - 0.110)
Total Phosphorus	mg/L	<0.02 (<0.02 - 0.04)	0.05 (0.02 - 0.06)	0.08 (0.03 - 0.13)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 204)	<50 (<50 - 101)	<50 (<50 - 636)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	2 (<1 - 4)
Copper	µg/L	<1 (<1 - 5)	<1 (<1 - <1)	3 (2 - 12)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	14 (<10 - 30)	<10 (<10 - 14)	14 (<10 - 71)
Flow	m ³ /s	0.195 (0.054 - 1.003)	0.065 (0.025 - 0.616)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ho Chung River in 2019

Parameter	Unit	<u>Ho Chung River</u>	
		PR1	PR2
Dissolved Oxygen	mg/L	7.7 (6.0 - 8.4)	8.3 (7.9 - 9.2)
pH		7.1 (6.8 - 7.8)	7.4 (6.8 - 7.6)
Suspended Solids	mg/L	6.3 (3.2 - 30.0)	3.5 (2.2 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	0.8 (0.3 - 9.0)	0.5 (0.1 - 1.2)
Chemical Oxygen Demand	mg/L	9 (4 - 30)	4 (<2 - 7)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	3 700 (370 - 130 000)	1 500 (500 - 5 500)
Faecal Coliforms	counts/ 100mL	10 000 (1 200 - 280 000)	5 900 (1 100 - 23 000)
Ammonia-Nitrogen	mg/L	0.385 (0.066 - 4.000)	0.050 (0.016 - 0.280)
Nitrate-Nitrogen	mg/L	0.440 (0.300 - 0.720)	0.355 (0.260 - 0.550)
Total Kjeldahl Nitrogen	mg/L	0.61 (0.22 - 4.90)	0.28 (0.16 - 1.20)
Orthophosphate Phosphorus	mg/L	0.039 (<0.002 - 0.450)	0.011 (<0.002 - 0.033)
Total Phosphorus	mg/L	0.06 (<0.02 - 0.56)	0.03 (<0.02 - 0.04)
Sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 670)	97 (<50 - 309)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 2)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 3)	<1 (<1 - <1)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	10 (<10 - 41)	<10 (<10 - 18)
Flow	m ³ /s	NM	1.575 (0.756 - 3.542)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sha Kok Mei Stream in 2019

Parameter	Unit	<u>Sha Kok Mei Stream</u>	
		PR5	PR6
Dissolved Oxygen	mg/L	7.4 (6.8 - 8.8)	8.3 (7.8 - 10.2)
pH		7.8 (7.1 - 8.0)	7.6 (7.1 - 7.7)
Suspended Solids	mg/L	2.4 (1.1 - 7.5)	2.0 (1.2 - 8.8)
5-Day Biochemical Oxygen Demand	mg/L	1.0 (<0.1 - 2.8)	1.0 (0.4 - 2.7)
Chemical Oxygen Demand	mg/L	5 (3 - 13)	5 (3 - 18)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.0)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	370 (<10 - 50 000)	12 000 (3 700 - 41 000)
Faecal Coliforms	counts/ 100mL	650 (<10 - 73 000)	23 000 (7 000 - 110 000)
Ammonia-Nitrogen	mg/L	0.210 (0.029 - 0.390)	0.160 (0.038 - 0.390)
Nitrate-Nitrogen	mg/L	0.855 (0.440 - 1.100)	1.650 (1.100 - 2.900)
Total Kjeldahl Nitrogen	mg/L	0.68 (0.39 - 1.10)	0.52 (0.27 - 1.30)
Orthophosphate Phosphorus	mg/L	0.041 (0.003 - 0.082)	0.069 (0.012 - 0.130)
Total Phosphorus	mg/L	0.08 (0.03 - 0.15)	0.09 (0.06 - 0.15)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 339)	56 (<50 - 320)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 19)	<10 (<10 - 22)
Flow	m ³ /s	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Chung Hau Stream in 2019

Parameter	Unit	Tai Chung Hau Stream	
		PR7	PR8
Dissolved Oxygen	mg/L	8.4 (7.6 - 10.4)	8.0 (7.6 - 8.8)
pH		7.7 (7.2 - 8.9)	8.1 (7.2 - 9.9)
Suspended Solids	mg/L	2.8 (1.1 - 8.4)	2.9 (1.5 - 12.0)
5-Day Biochemical Oxygen Demand	mg/L	1.3 (0.7 - 3.0)	2.9 (1.1 - 6.7)
Chemical Oxygen Demand	mg/L	7 (3 - 19)	6 (<2 - 49)
Oil & Grease	mg/L	<0.5 (<0.5 - 2.2)	<0.5 (<0.5 - 1.4)
<i>E. coli</i>	counts/ 100mL	9 600 (1 600 - 51 000)	4 600 (800 - 20 000)
Faecal Coliforms	counts/ 100mL	31 000 (9 400 - 100 000)	37 000 (9 300 - 110 000)
Ammonia-Nitrogen	mg/L	0.056 (0.015 - 0.220)	0.045 (0.014 - 0.092)
Nitrate-Nitrogen	mg/L	0.580 (0.360 - 1.500)	0.655 (0.530 - 2.000)
Total Kjeldahl Nitrogen	mg/L	0.37 (0.23 - 1.30)	0.46 (0.14 - 1.20)
Orthophosphate Phosphorus	mg/L	0.023 (<0.002 - 0.076)	0.028 (<0.002 - 0.086)
Total Phosphorus	mg/L	0.04 (0.02 - 0.11)	0.06 (<0.02 - 0.26)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	51 (<50 - 431)	75 (<50 - 354)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 6)
Copper	µg/L	<1 (<1 - 3)	<1 (<1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 22)	10 (<10 - 16)
Flow	m ³ /s	0.803 (0.135 - 1.280)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tseng Lan Shue Stream in 2019

Parameter	Unit	Tseng Lan Shue Stream		
		JR3	JR6	JR11
Dissolved Oxygen	mg/L	6.5 (2.8 - 7.5)	7.9 (7.2 - 8.4)	8.8 (7.9 - 10.5)
pH		7.1 (6.9 - 7.3)	7.3 (7.1 - 7.8)	7.1 (6.9 - 7.7)
Suspended Solids	mg/L	6.4 (3.5 - 250.0)	4.9 (2.4 - 16.0)	3.0 (1.4 - 9.1)
5-Day Biochemical Oxygen Demand	mg/L	9.3 (3.2 - 62.0)	5.0 (1.8 - 9.6)	1.0 (0.6 - 1.2)
Chemical Oxygen Demand	mg/L	13 (5 - 47)	11 (6 - 34)	8 (5 - 11)
Oil & Grease	mg/L	0.6 (<0.5 - 1.8)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	73 000 (41 000 - 190 000)	25 000 (5 900 - 66 000)	890 (130 - 11 000)
Faecal Coliforms	counts/ 100mL	120 000 (81 000 - 360 000)	47 000 (16 000 - 93 000)	2 800 (540 - 31 000)
Ammonia-Nitrogen	mg/L	3.950 (0.940 - 9.500)	0.275 (0.092 - 0.430)	0.078 (0.024 - 0.280)
Nitrate-Nitrogen	mg/L	1.400 (0.018 - 2.000)	2.050 (1.100 - 3.600)	2.550 (1.100 - 5.600)
Total Kjeldahl Nitrogen	mg/L	5.35 (1.80 - 17.00)	0.94 (0.60 - 2.20)	0.54 (0.20 - 1.80)
Orthophosphate Phosphorus	mg/L	0.385 (0.110 - 0.820)	0.340 (0.130 - 0.490)	0.300 (0.026 - 0.620)
Total Phosphorus	mg/L	0.50 (0.19 - 1.10)	0.45 (0.21 - 0.62)	0.33 (0.11 - 0.71)
Sulphide	mg/L	<0.02 (<0.02 - 0.15)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	63 (<50 - 249)	57 (<50 - 260)	<50 (<50 - 439)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	2 (1 - 4)	2 (1 - 5)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 3)	<1 (<1 - <1)
Zinc	µg/L	15 (11 - 34)	17 (<10 - 27)	<10 (<10 - 20)
Flow	m ³ /s	NM	NM	0.861 (0.192 - 2.652)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Indus in 2019

Parameter	Unit	River Indus		
		IN1	IN2	IN3
Dissolved Oxygen	mg/L	5.9 (3.3 - 7.6)	6.7 (5.8 - 8.7)	8.1 (7.6 - 9.0)
pH		7.0 (6.8 - 7.4)	7.3 (6.9 - 7.5)	7.6 (7.1 - 7.9)
Suspended Solids	mg/L	13.5 (3.7 - 140.0)	6.1 (3.5 - 9.6)	5.7 (2.3 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	3.7 (1.2 - 8.6)	1.9 (1.2 - 8.7)	0.6 (0.5 - 1.3)
Chemical Oxygen Demand	mg/L	22 (13 - 43)	9 (5 - 16)	8 (<2 - 14)
Oil & Grease	mg/L	<0.5 (<0.5 - 2.2)	<0.5 (<0.5 - 1.2)	<0.5 (<0.5 - 1.1)
<i>E. coli</i>	counts/ 100mL	16 000 (1 100 - 150 000)	2 700 (470 - 82 000)	2 200 (490 - 14 000)
Faecal Coliforms	counts/ 100mL	48 000 (3 200 - 470 000)	13 000 (4 300 - 200 000)	6 100 (1 100 - 31 000)
Ammonia-Nitrogen	mg/L	1.200 (0.220 - 2.300)	0.565 (0.330 - 1.000)	0.071 (0.037 - 0.140)
Nitrate-Nitrogen	mg/L	2.150 (1.200 - 5.500)	1.005 (0.620 - 1.600)	0.665 (0.350 - 1.400)
Total Kjeldahl Nitrogen	mg/L	2.95 (1.90 - 5.60)	1.30 (0.50 - 3.60)	0.37 (0.07 - 1.50)
Orthophosphate Phosphorus	mg/L	0.190 (0.076 - 0.830)	0.066 (0.039 - 0.100)	0.063 (0.040 - 0.110)
Total Phosphorus	mg/L	0.38 (0.18 - 0.94)	0.14 (0.08 - 0.21)	0.11 (0.07 - 0.19)
Sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	53 (<50 - 145)	<50 (<50 - 183)	70 (<50 - 127)
Cadmium	µg/L	<0.1 (<0.1 - 0.5)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	2 (1 - 5)	1 (<1 - 7)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	16 (<10 - 31)	10 (<10 - 29)	<10 (<10 - 18)
Flow	m ³ /s	10.626 (3.018 - 49.280)	NM	0.113 (0.015 - 0.220)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Beas in 2019

Parameter	Unit	River Beas		
		RB1	RB2	RB3
Dissolved Oxygen	mg/L	9.3 (8.5 - 12.7)	7.6 (6.8 - 9.1)	7.1 (4.6 - 9.7)
pH		7.8 (7.5 - 9.1)	7.3 (7.0 - 7.6)	7.2 (7.0 - 7.9)
Suspended Solids	mg/L	4.1 (1.1 - 8.7)	5.5 (3.6 - 15.0)	12.1 (5.3 - 28.0)
-Day Biochemical Oxygen Demand	mg/L	1.7 (0.3 - 2.7)	3.2 (0.3 - 17.0)	3.9 (0.3 - 36.0)
Chemical Oxygen Demand	mg/L	9 (5 - 15)	11 (5 - 19)	16 (12 - 48)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.9)	<0.5 (<0.5 - 0.9)	<0.5 (<0.5 - 3.3)
<i>E. coli</i>	counts/ 100mL	3 000 (570 - 16 000)	3 500 (300 - 29 000)	17 000 (1 800 - 55 000)
Faecal Coliforms	counts/ 100mL	10 000 (1 400 - 68 000)	8 500 (480 - 45 000)	48 000 (4 400 - 340 000)
Ammonia-Nitrogen	mg/L	0.135 (0.078 - 0.450)	1.150 (0.380 - 1.900)	1.050 (0.570 - 2.900)
Nitrate-Nitrogen	mg/L	0.850 (0.480 - 1.400)	0.635 (0.360 - 1.600)	0.955 (0.170 - 2.300)
Total Kjeldahl Nitrogen	mg/L	0.63 (0.32 - 3.90)	2.00 (0.89 - 4.80)	2.75 (1.30 - 4.70)
Orthophosphate Phosphorus	mg/L	0.170 (0.010 - 0.220)	0.115 (0.004 - 0.190)	0.135 (<0.002 - 0.280)
Total Phosphorus	mg/L	0.25 (0.13 - 0.30)	0.27 (0.14 - 0.37)	0.32 (0.21 - 0.64)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	<50 (<50 - 165)	<50 (<50 - 327)	<50 (<50 - 142)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 10)	1 (<1 - 2)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 21)	11 (<10 - 26)	17 (<10 - 59)
Flow	m ³ /s	0.135 (0.062 - 1.524)	0.301 (0.013 - 6.734)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Ganges in 2019

Parameter	Unit	River Ganges		
		GR1	GR2	GR3
Dissolved Oxygen	mg/L	7.6 (4.7 - 8.7)	5.4 (2.6 - 7.0)	7.8 (7.3 - 9.2)
pH		7.2 (7.1 - 7.5)	7.3 (7.0 - 7.7)	7.5 (6.9 - 7.6)
Suspended Solids	mg/L	12.0 (6.8 - 19.0)	10.0 (8.0 - 25.0)	5.6 (1.4 - 40.0)
5-Day Biochemical Oxygen Demand	mg/L	5.4 (0.4 - 12.0)	3.5 (0.2 - 9.5)	0.4 (0.1 - 1.8)
Chemical Oxygen Demand	mg/L	17 (13 - 25)	16 (11 - 25)	5 (<2 - 9)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 2.1)	<0.5 (<0.5 - 9.5)
<i>E. coli</i>	counts/ 100mL	4 600 (580 - 13 000)	4 900 (800 - 51 000)	370 (15 - 2 400)
Faecal Coliforms	counts/ 100mL	12 000 (4 900 - 25 000)	23 000 (3 600 - 330 000)	2 600 (390 - 90 000)
Ammonia-Nitrogen	mg/L	4.300 (0.440 - 21.000)	7.400 (0.300 - 31.000)	0.073 (<0.005 - 0.360)
Nitrate-Nitrogen	mg/L	1.020 (0.680 - 2.300)	0.600 (0.055 - 1.500)	0.255 (0.150 - 0.550)
Total Kjeldahl Nitrogen	mg/L	5.40 (0.80 - 21.00)	9.15 (1.40 - 34.00)	0.30 (0.06 - 0.98)
Orthophosphate Phosphorus	mg/L	0.555 (0.120 - 2.100)	0.230 (<0.002 - 0.620)	<0.002 (<0.002 - 0.018)
Total Phosphorus	mg/L	0.72 (0.39 - 2.70)	0.47 (0.18 - 2.00)	<0.02 (<0.02 - 0.05)
Sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - 0.05)
Aluminium	µg/L	<50 (<50 - 222)	66 (<50 - 222)	54 (<50 - 249)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	7 (2 - 20)	4 (<1 - 9)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - 2)	1 (<1 - 25)	<1 (<1 - <1)
Zinc	µg/L	15 (<10 - 30)	18 (<10 - 29)	<10 (<10 - 25)
Flow	m ³ /s	0.067 (0.012 - 0.266)	0.114 (0.020 - 0.289)	0.055 (0.020 - 0.202)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2019 (Part 1 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL1	YL2
Dissolved Oxygen	mg/L	4.8 (2.2 - 7.6)	6.9 (4.4 - 8.1)
pH		7.1 (7.0 - 7.6)	7.3 (7.2 - 11.4)
Suspended Solids	mg/L	38.5 (7.8 - 76.0)	5.1 (4.1 - 56.0)
5-Day Biochemical Oxygen Demand	mg/L	11.3 (4.0 - 54.0)	6.8 (2.9 - 29.0)
Chemical Oxygen Demand	mg/L	34 (12 - 160)	16 (11 - 24)
Oil & Grease	mg/L	0.7 (<0.5 - 9.1)	<0.5 (<0.5 - 0.7)
<i>E. coli</i>	counts/ 100mL	120 000 (23 000 - 490 000)	26 000 (1 - 560 000)
Faecal Coliforms	counts/ 100mL	290 000 (65 000 - 1 700 000)	47 000 (1 - 660 000)
Ammonia-Nitrogen	mg/L	5.350 (0.960 - 15.000)	3.750 (0.940 - 5.800)
Nitrate-Nitrogen	mg/L	0.530 (<0.002 - 1.600)	0.705 (0.180 - 1.800)
Total Kjeldahl Nitrogen	mg/L	6.65 (1.50 - 19.00)	5.00 (1.60 - 7.00)
Orthophosphate Phosphorus	mg/L	0.515 (0.170 - 2.200)	0.245 (0.140 - 0.470)
Total Phosphorus	mg/L	0.72 (0.30 - 2.80)	0.38 (0.24 - 0.62)
Sulphide	mg/L	0.03 (<0.02 - 0.21)	<0.02 (<0.02 - 0.08)
Aluminium	µg/L	55 (<50 - 922)	58 (<50 - 529)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 11)
Copper	µg/L	3 (2 - 6)	2 (<1 - 19)
Lead	µg/L	<1 (<1 - 4)	<1 (<1 - <1)
Zinc	µg/L	13 (<10 - 29)	11 (<10 - 33)
Flow	m ³ /s	0.522 (0.131 - 2.873)	0.024 (0.009 - 0.111)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2019 (Part 2 of 2)

Parameter	Unit	<u>Yuen Long Creek</u>	
		YL3	YL4
Dissolved Oxygen	mg/L	2.8 (1.1 - 6.2)	3.4 (1.7 - 6.3)
pH		7.2 (6.9 - 7.7)	7.2 (7.0 - 7.5)
Suspended Solids	mg/L	35.5 (18.0 - 170.0)	54.0 (11.0 - 87.0)
5-Day Biochemical Oxygen Demand	mg/L	30.5 (7.0 - 86.0)	92.0 (8.8 - 170.0)
Chemical Oxygen Demand	mg/L	54 (16 - 180)	120 (7 - 210)
Oil & Grease	mg/L	1.9 (<0.5 - 4.3)	4.8 (0.9 - 7.6)
<i>E. coli</i>	counts/ 100mL	500 000 (90 000 - 1 700 000)	1 700 000 (260 000 - 4 200 000)
Faecal Coliforms	counts/ 100mL	1 200 000 (160 000 - 4 900 000)	3 700 000 (820 000 - 8 900 000)
Ammonia-Nitrogen	mg/L	5.600 (1.400 - 15.000)	5.400 (0.500 - 10.000)
Nitrate-Nitrogen	mg/L	0.006 (<0.002 - 0.690)	0.006 (<0.002 - 0.380)
Total Kjeldahl Nitrogen	mg/L	8.60 (2.50 - 41.00)	9.35 (1.10 - 14.00)
Orthophosphate Phosphorus	mg/L	0.465 (0.160 - 1.700)	0.265 (0.027 - 0.600)
Total Phosphorus	mg/L	0.82 (0.33 - 4.50)	0.86 (0.13 - 1.40)
Sulphide	mg/L	0.07 (<0.02 - 0.16)	0.13 (<0.02 - 0.32)
Aluminium	µg/L	<50 (<50 - 768)	64 (<50 - 264)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	4 (1 - 7)	3 (2 - 7)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 2)
Zinc	µg/L	17 (<10 - 51)	18 (<10 - 45)
Flow	m ³ /s	1.065 (0.258 - 3.768)	0.268 (0.099 - 0.656)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kam Tin River in 2019

Parameter	Unit	<u>Kam Tin River</u>	
		KT1	KT2
Dissolved Oxygen	mg/L	5.9 (3.6 - 7.0)	4.0 (1.8 - 7.6)
pH		7.5 (7.1 - 7.8)	7.4 (7.2 - 7.7)
Suspended Solids	mg/L	12.0 (6.5 - 24.0)	23.5 (12.0 - 50.0)
5-Day Biochemical Oxygen Demand	mg/L	8.8 (3.7 - 19.0)	16.0 (4.7 - 37.0)
Chemical Oxygen Demand	mg/L	17 (12 - 29)	31 (4 - 100)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.9)	<0.5 (<0.5 - 2.3)
<i>E. coli</i>	counts/ 100mL	56 000 (19 000 - 280 000)	51 000 (14 000 - 340 000)
Faecal Coliforms	counts/ 100mL	160 000 (54 000 - 400 000)	110 000 (36 000 - 490 000)
Ammonia-Nitrogen	mg/L	3.450 (1.000 - 13.000)	7.050 (0.830 - 10.000)
Nitrate-Nitrogen	mg/L	0.760 (0.280 - 1.700)	0.150 (<0.002 - 0.900)
Total Kjeldahl Nitrogen	mg/L	4.05 (1.80 - 15.00)	8.40 (1.50 - 14.00)
Orthophosphate Phosphorus	mg/L	0.475 (0.300 - 1.700)	0.785 (0.190 - 1.700)
Total Phosphorus	mg/L	0.66 (0.46 - 2.00)	1.25 (0.32 - 2.40)
Sulphide	mg/L	<0.02 (<0.02 - 0.06)	0.04 (<0.02 - 0.27)
Aluminium	µg/L	<50 (<50 - 409)	<50 (<50 - 442)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (2 - 7)	2 (<1 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	14 (<10 - 26)	12 (<10 - 34)
Flow	m ³ /s	1.411 (0.324 - 8.979)	1.394 (0.238 - 4.454)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tin Shui Wai Nullah and Fairview Park Nullah in 2019

Parameter	Unit	Tin Shui Wai Nullah		Fairview Park Nullah
		TSR1	TSR2	FVR1
Dissolved Oxygen	mg/L	7.0 (3.3 - 12.1)	9.1 (8.2 - 11.1)	4.9 (3.2 - 9.6)
pH		7.5 (7.0 - 9.2)	8.2 (7.5 - 8.6)	7.3 (7.1 - 8.2)
Suspended Solids	mg/L	9.4 (2.9 - 25.0)	3.5 (2.2 - 36.0)	29.5 (17.0 - 100.0)
5-Day Biochemical Oxygen Demand	mg/L	4.1 (2.9 - 6.8)	1.5 (0.5 - 15.0)	6.3 (2.8 - 9.2)
Chemical Oxygen Demand	mg/L	11 (7 - 19)	7 (4 - 40)	22 (3 - 45)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	51 000 (9 800 - 330 000)	31 000 (11 000 - 400 000)	13 000 (2 800 - 1 800 000)
Faecal Coliforms	counts/ 100mL	140 000 (43 000 - 680 000)	59 000 (23 000 - 770 000)	39 000 (11 000 - 1 800 000)
Ammonia-Nitrogen	mg/L	1.550 (0.660 - 4.000)	0.310 (0.140 - 3.600)	1.400 (0.370 - 2.900)
Nitrate-Nitrogen	mg/L	0.620 (0.390 - 1.300)	0.770 (0.340 - 1.200)	0.760 (0.270 - 1.600)
Total Kjeldahl Nitrogen	mg/L	2.40 (1.50 - 5.10)	0.90 (0.29 - 4.40)	2.45 (1.20 - 4.50)
Orthophosphate Phosphorus	mg/L	0.125 (<0.002 - 0.320)	0.051 (0.002 - 0.140)	0.285 (0.110 - 0.460)
Total Phosphorus	mg/L	0.20 (0.12 - 0.51)	0.09 (0.04 - 0.45)	0.46 (0.26 - 0.70)
Sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.08)	0.02 (<0.02 - 0.04)
Aluminium	µg/L	<50 (<50 - 173)	125 (<50 - 235)	<50 (<50 - 213)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 2)
Copper	µg/L	2 (<1 - 4)	1 (<1 - 4)	2 (<1 - 3)
Lead	µg/L	<1 (<1 - 4)	<1 (<1 - <1)	<1 (<1 - 3)
Zinc	µg/L	13 (<10 - 40)	<10 (<10 - 55)	19 (10 - 32)
Flow	m ³ /s	NM	0.068 (0.010 - 0.306)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ha Pak Nai Stream, Pak Nai Stream and Sheung Pak Nai Stream in 2019

Parameter	Unit	Ha Pak Nai Stream	Pak Nai Stream	Sheung Pak Nai Stream
		DB1	DB3	DB5
Dissolved Oxygen	mg/L	8.7 (7.8 - 10.0)	8.4 (7.4 - 9.7)	8.1 (7.4 - 9.4)
pH		7.5 (6.5 - 7.8)	7.0 (6.1 - 7.4)	7.1 (6.4 - 7.3)
Suspended Solids	mg/L	2.2 (1.2 - 27.0)	3.8 (1.9 - 15.0)	6.5 (2.4 - 17.0)
5-Day Biochemical Oxygen Demand	mg/L	<0.1 (<0.1 - 0.4)	0.2 (<0.1 - 1.4)	1.1 (0.2 - 5.4)
Chemical Oxygen Demand	mg/L	3 (<2 - 6)	3 (<2 - 6)	7 (<2 - 10)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 1.2)	<0.5 (<0.5 - 1.0)
<i>E. coli</i>	counts/ 100mL	59 (5 - 340)	240 (74 - 900)	5 400 (370 - 180 000)
Faecal Coliforms	counts/ 100mL	490 (120 - 5 800)	1 700 (510 - 6 000)	9 600 (1 100 - 200 000)
Ammonia-Nitrogen	mg/L	0.017 (<0.005 - 0.040)	0.020 (0.009 - 0.037)	0.098 (0.021 - 1.400)
Nitrate-Nitrogen	mg/L	0.440 (0.280 - 0.680)	0.355 (0.220 - 0.530)	0.250 (0.180 - 0.390)
Total Kjeldahl Nitrogen	mg/L	0.15 (<0.05 - 0.39)	0.13 (0.07 - 0.23)	0.42 (0.11 - 3.00)
Orthophosphate Phosphorus	mg/L	0.005 (<0.002 - 0.011)	0.003 (<0.002 - 0.018)	0.013 (<0.002 - 0.290)
Total Phosphorus	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	0.04 (<0.02 - 0.45)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	359 (54 - 820)	300 (65 - 747)	278 (54 - 612)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - <1)	<1 (<1 - 3)	<1 (<1 - 3)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - 2)
Zinc	µg/L	<10 (<10 - 27)	<10 (<10 - 29)	<10 (<10 - 17)
Flow	m ³ /s	0.034 (0.002 - 0.181)	0.063 (0.029 - 0.108)	0.089 (0.012 - 0.252)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream in 2019

Parameter	Unit	<u>Ngau Hom Sha Stream</u>	<u>Tai Shui Hang Stream</u>	<u>Tsang Kok Stream</u>
		DB6	DB2	DB8
Dissolved Oxygen	mg/L	8.4 (7.4 - 10.1)	8.9 (7.9 - 10.2)	8.5 (6.7 - 13.4)
pH		6.9 (6.5 - 7.1)	7.7 (6.8 - 7.9)	7.4 (7.0 - 9.0)
Suspended Solids	mg/L	4.8 (1.8 - 15.0)	2.9 (1.9 - 18.0)	5.5 (1.0 - 12.0)
5-Day Biochemical Oxygen Demand	mg/L	0.6 (0.3 - 1.3)	0.2 (<0.1 - 0.9)	0.6 (0.2 - 1.4)
Chemical Oxygen Demand	mg/L	4 (<2 - 15)	3 (<2 - 6)	5 (2 - 19)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.1)	<0.5 (<0.5 - 1.4)	<0.5 (<0.5 - 1.4)
<i>E. coli</i>	counts/ 100mL	350 (90 - 1 900)	130 (29 - 800)	230 (70 - 800)
Faecal Coliforms	counts/ 100mL	2 100 (370 - 4 900)	680 (48 - 3 200)	1 900 (140 - 6 300)
Ammonia-Nitrogen	mg/L	0.104 (0.027 - 0.320)	0.015 (0.006 - 0.023)	0.047 (0.018 - 9.000)
Nitrate-Nitrogen	mg/L	0.165 (0.047 - 0.340)	0.185 (0.120 - 0.400)	1.600 (0.920 - 2.500)
Total Kjeldahl Nitrogen	mg/L	0.27 (0.12 - 0.51)	0.13 (<0.05 - 0.63)	0.36 (0.15 - 13.00)
Orthophosphate Phosphorus	mg/L	0.032 (0.008 - 0.110)	<0.002 (<0.002 - 0.007)	0.006 (<0.002 - 0.040)
Total Phosphorus	mg/L	0.08 (0.03 - 0.31)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.14)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	76 (<50 - 349)	215 (56 - 676)	183 (53 - 311)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	<1 (<1 - 12)	<1 (<1 - 3)	<1 (<1 - 4)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - 2)
Zinc	µg/L	12 (<10 - 31)	<10 (<10 - 29)	10 (<10 - 17)
Flow	m ³ /s	0.044 (0.012 - 0.240)	0.262 (0.100 - 0.624)	0.025 (0.012 - 0.082)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2019 (Part 1 of 2)

Parameter	Unit	<u>Mui Wo River</u>		
		MW1	MW2	MW3
Dissolved Oxygen	mg/L	8.3 (7.2 - 9.6)	7.9 (7.2 - 10.8)	8.6 (7.9 - 10.2)
pH		7.6 (7.0 - 8.0)	7.6 (7.0 - 7.9)	7.0 (6.6 - 7.4)
Suspended Solids	mg/L	3.5 (0.7 - 12.0)	7.1 (3.4 - 20.0)	1.6 (<0.5 - 23.0)
5-Day Biochemical Oxygen Demand	mg/L	0.3 (<0.1 - 1.5)	0.9 (0.5 - 1.9)	0.3 (<0.1 - 1.0)
Chemical Oxygen Demand	mg/L	5 (2 - 10)	11 (5 - 18)	3 (<2 - 7)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 1.4)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	840 (170 - 3 000)	2 300 (690 - 6 500)	210 (63 - 4 200)
Faecal Coliforms	counts/ 100mL	4 800 (1 200 - 10 000)	10 000 (3 900 - 52 000)	4 000 (900 - 9 200)
Ammonia-Nitrogen	mg/L	0.062 (0.022 - 0.160)	0.295 (0.046 - 0.770)	0.015 (0.005 - 0.031)
Nitrate-Nitrogen	mg/L	0.350 (0.220 - 0.640)	0.260 (<0.002 - 0.380)	0.320 (0.150 - 0.750)
Total Kjeldahl Nitrogen	mg/L	0.19 (0.10 - 0.49)	0.46 (0.29 - 1.40)	0.13 (<0.05 - 0.42)
Orthophosphate Phosphorus	mg/L	0.061 (<0.002 - 0.120)	0.037 (0.002 - 0.120)	0.040 (<0.002 - 0.072)
Total Phosphorus	mg/L	0.08 (0.05 - 0.15)	0.10 (0.05 - 0.18)	0.05 (0.03 - 0.08)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	69 (<50 - 347)	60 (<50 - 443)	79 (<50 - 254)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	1 (<1 - 2)	<1 (<1 - 3)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 4)	1 (<1 - 6)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - 5)	<1 (<1 - 5)	<1 (<1 - 1)
Zinc	µg/L	14 (<10 - 23)	14 (<10 - 23)	13 (<10 - 18)
Flow	m ³ /s	0.114 (0.030 - 0.854)	NM	0.079 (0.010 - 2.509)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2019 (Part 2 of 2)

Parameter	Unit	<u>Mui Wo River</u>	
		MW4	MW5
Dissolved Oxygen	mg/L	7.4 (5.1 - 8.2)	7.7 (5.8 - 9.3)
pH		7.1 (6.6 - 8.1)	7.4 (7.0 - 7.6)
Suspended Solids	mg/L	8.5 (2.2 - 49.0)	6.5 (1.9 - 79.0)
5-Day Biochemical Oxygen Demand	mg/L	0.7 (0.3 - 1.5)	1.4 (0.6 - 2.7)
Chemical Oxygen Demand	mg/L	10 (7 - 16)	9 (4 - 21)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
<i>E. coli</i>	counts/ 100mL	610 (50 - 3 300)	4 000 (1 300 - 12 000)
Faecal Coliforms	counts/ 100mL	3 000 (190 - 18 000)	25 000 (3 500 - 110 000)
Ammonia-Nitrogen	mg/L	0.215 (0.049 - 0.610)	0.490 (0.065 - 1.300)
Nitrate-Nitrogen	mg/L	0.280 (0.130 - 0.490)	0.190 (<0.002 - 0.330)
Total Kjeldahl Nitrogen	mg/L	0.40 (0.16 - 0.70)	0.65 (0.30 - 1.70)
Orthophosphate Phosphorus	mg/L	0.039 (<0.002 - 0.060)	0.043 (0.002 - 0.170)
Total Phosphorus	mg/L	0.08 (0.05 - 0.11)	0.09 (0.04 - 0.34)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	<50 (<50 - 438)	64 (<50 - 412)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 3)	1 (<1 - 2)
Copper	µg/L	4 (<1 - 5)	2 (<1 - 6)
Lead	µg/L	<1 (<1 - 6)	<1 (<1 - 12)
Zinc	µg/L	14 (<10 - 30)	16 (<10 - 25)
Flow	m ³ /s	0.532 (0.185 - 0.870)	0.108 (0.041 - 0.720)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tung Chung River in 2019

Parameter	Unit	Tung Chung River		
		TC1	TC2	TC3
Dissolved Oxygen	mg/L	7.7 (7.3 - 9.3)	8.3 (7.7 - 9.8)	8.2 (7.4 - 8.9)
pH		7.1 (6.7 - 7.5)	7.7 (7.0 - 8.6)	7.9 (7.3 - 8.4)
Suspended Solids	mg/L	1.0 (0.6 - 14.0)	2.2 (1.1 - 3.8)	3.9 (0.6 - 18.0)
5-Day Biochemical Oxygen Demand	mg/L	0.1 (<0.1 - 0.7)	0.6 (0.4 - 1.2)	5.8 (0.6 - 17.0)
Chemical Oxygen Demand	mg/L	2 (<2 - 6)	5 (3 - 12)	10 (3 - 18)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.2)
<i>E. coli</i>	counts/ 100mL	98 (17 - 3 900)	760 (90 - 6 400)	20 000 (3 200 - 59 000)
Faecal Coliforms	counts/ 100mL	1 800 (230 - 19 000)	5 700 (480 - 39 000)	59 000 (12 000 - 120 000)
Ammonia-Nitrogen	mg/L	0.017 (0.009 - 0.027)	0.034 (0.018 - 0.074)	1.500 (0.680 - 4.100)
Nitrate-Nitrogen	mg/L	0.075 (0.014 - 0.500)	0.074 (0.006 - 0.310)	0.115 (0.093 - 1.200)
Total Kjeldahl Nitrogen	mg/L	0.07 (<0.05 - 0.22)	0.20 (<0.05 - 0.29)	2.10 (1.10 - 4.90)
Orthophosphate Phosphorus	mg/L	<0.002 (<0.002 - 0.010)	0.010 (<0.002 - 0.021)	0.054 (0.004 - 0.130)
Total Phosphorus	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.04)	0.12 (<0.02 - 0.24)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 345)	<50 (<50 - 270)	<50 (<50 - 177)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	1 (<1 - 2)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 25)	<10 (<10 - 13)	<10 (<10 - 21)
Flow	m ³ /s	0.073 (0.021 - 2.506)	0.167 (0.030 - 0.712)	0.061 (0.020 - 1.078)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2019 (Part 1 of 2)

Parameter	Unit	Tuen Mun River		
		TN1	TN2	TN3
Dissolved Oxygen	mg/L	4.3 (2.2 - 5.5)	7.7 (6.2 - 9.3)	5.0 (3.6 - 6.5)
pH		7.7 (7.5 - 7.9)	7.8 (7.1 - 7.9)	7.5 (7.3 - 8.0)
Suspended Solids	mg/L	11.5 (6.7 - 21.0)	4.6 (2.1 - 10.0)	5.1 (2.8 - 24.0)
5-Day Biochemical Oxygen Demand	mg/L	18.0 (11.0 - 25.0)	2.8 (0.6 - 12.0)	3.3 (1.1 - 21.0)
Chemical Oxygen Demand	mg/L	26 (18 - 33)	11 (5 - 17)	11 (7 - 42)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.3)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	84 000 (46 000 - 170 000)	17 000 (3 500 - 210 000)	5 700 (370 - 28 000)
Faecal Coliforms	counts/ 100mL	360 000 (110 000 - 1 500 000)	43 000 (9 300 - 460 000)	58 000 (2 800 - 950 000)
Ammonia-Nitrogen	mg/L	6.600 (5.200 - 8.800)	0.955 (0.380 - 6.200)	0.465 (0.100 - 0.920)
Nitrate-Nitrogen	mg/L	0.310 (<0.002 - 0.800)	1.100 (0.550 - 2.400)	0.395 (0.003 - 0.790)
Total Kjeldahl Nitrogen	mg/L	8.45 (6.50 - 11.00)	1.80 (0.70 - 7.70)	1.10 (0.67 - 1.80)
Orthophosphate Phosphorus	mg/L	0.465 (0.280 - 0.680)	0.145 (0.032 - 0.520)	0.032 (0.010 - 0.073)
Total Phosphorus	mg/L	0.70 (0.53 - 0.97)	0.20 (0.07 - 0.68)	0.08 (0.06 - 0.13)
Sulphide	mg/L	0.03 (<0.02 - 0.07)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.11)
Aluminium	µg/L	<50 (<50 - 79)	100 (<50 - 280)	<50 (<50 - 107)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 2)	2 (<1 - 4)
Copper	µg/L	2 (1 - 4)	<1 (<1 - 3)	4 (2 - 6)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 3)	<1 (<1 - <1)
Zinc	µg/L	14 (<10 - 26)	14 (<10 - 26)	14 (<10 - 21)
Flow	m ³ /s	0.158 (0.079 - 0.317)	0.038 (0.012 - 0.274)	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2019 (Part 2 of 2)

Parameter	Unit	Tuen Mun River		
		TN4	TN5	TN6
Dissolved Oxygen	mg/L	4.7 (3.7 - 13.1)	5.4 (3.3 - 10.2)	5.1 (2.9 - 6.3)
pH		7.4 (7.2 - 8.7)	7.5 (7.2 - 8.4)	7.2 (6.7 - 7.7)
Suspended Solids	mg/L	4.9 (2.7 - 19.0)	5.2 (2.8 - 10.0)	4.1 (2.0 - 8.8)
5-Day Biochemical Oxygen Demand	mg/L	1.9 (0.6 - 6.2)	2.3 (1.1 - 6.3)	2.1 (0.6 - 3.6)
Chemical Oxygen Demand	mg/L	12 (6 - 20)	11 (6 - 18)	13 (5 - 20)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 800 (400 - 57 000)	5 400 (300 - 33 000)	4 600 (800 - 50 000)
Faecal Coliforms	counts/ 100mL	42 000 (3 300 - 490 000)	50 000 (6 000 - 2 100 000)	35 000 (7 000 - 180 000)
Ammonia-Nitrogen	mg/L	0.630 (0.068 - 1.100)	0.520 (0.260 - 0.940)	0.500 (0.280 - 0.730)
Nitrate-Nitrogen	mg/L	0.410 (0.110 - 0.750)	0.360 (0.140 - 0.730)	0.365 (0.140 - 0.550)
Total Kjeldahl Nitrogen	mg/L	1.15 (0.70 - 1.80)	1.10 (0.67 - 2.20)	1.20 (0.68 - 2.30)
Orthophosphate Phosphorus	mg/L	0.033 (0.005 - 0.066)	0.030 (0.004 - 0.066)	0.030 (<0.002 - 0.051)
Total Phosphorus	mg/L	0.08 (0.05 - 0.13)	0.08 (0.05 - 0.13)	0.08 (0.04 - 0.15)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	52 (<50 - 189)	52 (<50 - 134)	<50 (<50 - 121)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	1 (<1 - 4)	2 (<1 - 4)	2 (<1 - 4)
Copper	µg/L	4 (2 - 6)	4 (3 - 7)	4 (3 - 7)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	13 (<10 - 29)	13 (<10 - 19)	15 (10 - 29)
Flow	m ³ /s	NM	NM	NM

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
 4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
 5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Pai Min Kok Stream and Kau Wa Keng Stream in 2019

Parameter	Unit	Pai Min Kok Stream		Kau Wa Keng Stream
		AN1	AN2	KW3
Dissolved Oxygen	mg/L	8.3 (7.8 - 9.8)	8.7 (7.9 - 10.1)	8.1 (7.4 - 10.2)
pH		8.1 (7.6 - 9.8)	8.1 (7.5 - 9.4)	7.1 (6.8 - 7.4)
Suspended Solids	mg/L	4.4 (<0.5 - 48.0)	5.0 (0.9 - 14.0)	4.1 (1.0 - 19.0)
5-Day Biochemical Oxygen Demand	mg/L	0.8 (0.2 - 6.1)	0.6 (0.2 - 1.0)	2.6 (0.7 - 6.4)
Chemical Oxygen Demand	mg/L	8 (3 - 18)	7 (3 - 12)	10 (5 - 16)
Oil & Grease	mg/L	<0.5 (<0.5 - 2.1)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)
<i>E. coli</i>	counts/ 100mL	6 700 (20 - 370 000)	7 300 (100 - 810 000)	2 600 (6 - 170 000)
Faecal Coliforms	counts/ 100mL	22 000 (80 - 770 000)	12 000 (100 - 820 000)	4 600 (8 - 230 000)
Ammonia-Nitrogen	mg/L	0.061 (0.031 - 0.640)	0.020 (<0.005 - 0.064)	0.515 (0.130 - 1.400)
Nitrate-Nitrogen	mg/L	0.640 (0.230 - 0.990)	0.260 (0.005 - 0.650)	2.100 (0.570 - 2.900)
Total Kjeldahl Nitrogen	mg/L	0.53 (0.28 - 1.20)	0.28 (0.14 - 0.50)	1.01 (0.47 - 2.40)
Orthophosphate Phosphorus	mg/L	0.046 (0.025 - 0.094)	0.037 (0.013 - 0.058)	0.007 (<0.002 - 0.027)
Total Phosphorus	mg/L	0.07 (0.05 - 0.15)	0.06 (0.04 - 0.07)	0.07 (<0.02 - 0.11)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	146 (<50 - 430)	353 (<50 - 788)	<50 (<50 - 180)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	2.6 (0.4 - 6.5)
Chromium	µg/L	<1 (<1 - 5)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	3 (2 - 7)	2 (1 - 5)	2 (1 - 3)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	10 (<10 - 36)	13 (<10 - 25)	185 (53 - 317)
Flow	m ³ /s	NM	0.007 (0.002 - 0.043)	0.023 (0.009 - 0.223)

- Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
 2. Figures in brackets are annual ranges.
 3. NM indicates no measurement taken.
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Summary of water quality monitoring data for Sam Dip Tam Stream in 2019

Parameter	Unit	Sam Dip Tam Stream		
		TW1	TW2	TW3
Dissolved Oxygen	mg/L	8.0 (7.5 - 9.1)	8.3 (7.9 - 9.8)	9.0 (8.1 - 10.1)
pH		7.6 (7.5 - 8.0)	8.0 (7.9 - 8.3)	8.0 (7.7 - 8.4)
Suspended Solids	mg/L	2.4 (1.0 - 4.5)	1.8 (<0.5 - 3.0)	1.7 (0.7 - 4.0)
5-Day Biochemical Oxygen Demand	mg/L	0.9 (0.4 - 1.9)	0.8 (0.5 - 1.5)	0.8 (0.5 - 2.7)
Chemical Oxygen Demand	mg/L	3 (2 - 9)	5 (3 - 9)	6 (3 - 10)
Oil & Grease	mg/L	<0.5 (<0.5 - 1.6)	<0.5 (<0.5 - 0.9)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	1 200 (10 - 64 000)	1 300 (10 - 11 000)	3 100 (130 - 12 000)
Faecal Coliforms	counts/ 100mL	13 000 (100 - 110 000)	5 000 (10 - 40 000)	7 700 (180 - 28 000)
Ammonia-Nitrogen	mg/L	0.026 (<0.005 - 0.130)	0.098 (0.047 - 0.290)	0.120 (0.036 - 0.360)
Nitrate-Nitrogen	mg/L	0.690 (0.520 - 1.100)	1.100 (0.820 - 1.600)	1.200 (0.990 - 1.800)
Total Kjeldahl Nitrogen	mg/L	0.31 (0.08 - 0.67)	0.37 (0.10 - 0.70)	0.40 (0.18 - 1.00)
Orthophosphate Phosphorus	mg/L	0.040 (0.019 - 0.058)	0.100 (0.051 - 0.190)	0.115 (0.076 - 0.150)
Total Phosphorus	mg/L	0.06 (0.03 - 0.09)	0.13 (0.07 - 0.21)	0.15 (0.09 - 0.17)
Sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 130)	<50 (<50 - 145)	<50 (<50 - 92)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	1 (<1 - 2)	1 (<1 - 13)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	<10 (<10 - 36)	<10 (<10 - 19)	<10 (<10 - 11)
Flow	m ³ /s	NM	0.114 (0.019 - 0.421)	NM

- Notes:
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Summary of water quality monitoring data for Kai Tak River in 2019 (Part 1 of 2)

Parameter	Unit	Kai Tak River		
		KN1	KN2	KN3
Dissolved Oxygen	mg/L	4.8 (1.5 - 6.0)	6.1 (5.6 - 7.5)	6.6 (5.9 - 7.6)
pH		7.3 (7.2 - 7.5)	7.3 (7.1 - 7.6)	7.3 (7.1 - 7.5)
Suspended Solids	mg/L	4.2 (1.4 - 7.8)	4.9 (2.6 - 13.0)	6.3 (1.5 - 10.0)
5-Day Biochemical Oxygen Demand	mg/L	2.5 (1.4 - 5.0)	3.6 (1.8 - 6.4)	4.0 (2.3 - 6.8)
Chemical Oxygen Demand	mg/L	21 (11 - 35)	28 (13 - 36)	26 (14 - 36)
Oil & Grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.8)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	30 000 (1 100 - 670 000)	7 200 (2 100 - 220 000)	5 100 (2 000 - 20 000)
Faecal Coliforms	counts/ 100mL	73 000 (3 100 - 1 100 000)	17 000 (4 500 - 680 000)	15 000 (4 800 - 83 000)
Ammonia-Nitrogen	mg/L	1.850 (0.700 - 4.500)	1.750 (0.360 - 4.400)	1.650 (0.610 - 3.800)
Nitrate-Nitrogen	mg/L	2.350 (0.680 - 3.200)	3.150 (0.370 - 4.200)	3.450 (1.400 - 4.600)
Total Kjeldahl Nitrogen	mg/L	3.15 (1.80 - 7.40)	2.85 (1.80 - 6.70)	3.35 (1.50 - 6.10)
Orthophosphate Phosphorus	mg/L	1.000 (0.310 - 1.400)	1.200 (0.160 - 1.700)	1.050 (0.850 - 1.700)
Total Phosphorus	mg/L	1.15 (0.35 - 1.60)	1.35 (0.24 - 1.90)	1.25 (1.00 - 2.10)
Sulphide	mg/L	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	<50 (<50 - 97)	<50 (<50 - 103)	<50 (<50 - 92)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 2)	2 (1 - 2)	2 (<1 - 2)
Copper	µg/L	3 (3 - 5)	3 (2 - 4)	3 (3 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	18 (12 - 36)	23 (11 - 33)	23 (11 - 36)
Flow	m ³ /s	NM	NM	NM

- Notes:
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 3. NM indicates no measurement taken.
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Summary of water quality monitoring data for Kai Tak River in 2019 (Part 2 of 2)

Parameter	Unit	Kai Tak River		
		KN4	KN5	KN7
Dissolved Oxygen	mg/L	6.6 (5.7 - 7.6)	7.2 (7.0 - 8.0)	7.4 (6.8 - 8.0)
pH		7.3 (7.1 - 7.5)	7.3 (7.1 - 7.5)	7.1 (7.0 - 7.3)
Suspended Solids	mg/L	6.9 (4.4 - 13.0)	6.8 (4.9 - 16.0)	7.3 (3.6 - 13.0)
5-Day Biochemical Oxygen Demand	mg/L	4.5 (2.8 - 9.8)	5.3 (3.1 - 13.0)	5.4 (3.0 - 10.0)
Chemical Oxygen Demand	mg/L	30 (18 - 36)	26 (14 - 48)	24 (13 - 42)
Oil & Grease	mg/L	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
<i>E. coli</i>	counts/ 100mL	4 900 (1 400 - 35 000)	4 100 (2 100 - 19 000)	2 100 (520 - 25 000)
Faecal Coliforms	counts/ 100mL	12 000 (2 400 - 87 000)	10 000 (4 400 - 67 000)	3 800 (1 400 - 25 000)
Ammonia-Nitrogen	mg/L	1.450 (0.540 - 3.500)	1.070 (0.350 - 3.100)	1.005 (0.220 - 2.800)
Nitrate-Nitrogen	mg/L	3.850 (1.400 - 4.800)	4.250 (2.900 - 5.100)	4.300 (2.900 - 4.900)
Total Kjeldahl Nitrogen	mg/L	2.85 (1.40 - 5.50)	2.70 (1.20 - 5.50)	2.75 (1.10 - 5.10)
Orthophosphate Phosphorus	mg/L	1.000 (0.820 - 1.700)	1.040 (0.590 - 1.500)	1.000 (0.620 - 1.500)
Total Phosphorus	mg/L	1.20 (1.00 - 1.90)	1.30 (0.91 - 1.70)	1.20 (0.91 - 1.70)
Sulphide	mg/L	<0.02 (<0.02 - 0.18)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - <50)	<50 (<50 - 67)	<50 (<50 - 80)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	2 (<1 - 2)	2 (<1 - 2)	1 (<1 - 2)
Copper	µg/L	3 (3 - 4)	3 (3 - 4)	3 (2 - 4)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	22 (12 - 30)	24 (12 - 34)	24 (12 - 33)
Flow	m ³ /s	NM	13.966 (8.228 - 41.194)	9.659 (1.688 - 13.520)

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WQO compliance rates for rivers and streams in 2019

Watercourse	Water Quality Objectives (WQOs)					
	pH	5-Day Biochemical Oxygen Demand	Chemical Oxygen Demand	Dissolved Oxygen	Suspended Solids*	Overall Compliance Rate (%)
Eastern New Territories						
Shing Mun River	81%	93%	99%	100%	100%	95%
Lam Tsuen River	98%	93%	96%	98%	100%	97%
Tai Po River	100%	100%	100%	100%	100%	100%
Tai Po Kau Stream	100%	100%	92%	100%	100%	98%
Shan Liu Stream	100%	100%	100%	100%	100%	100%
Tung Tze Stream	100%	100%	100%	83%	100%	97%
Ho Chung River	100%	92%	100%	100%	100%	98%
Sha Kok Mei Stream	100%	100%	100%	100%	100%	100%
Tai Chung Hau Stream	92%	96%	96%	100%	100%	97%
Tseng Lan Shue Stream	100%	56%	92%	97%	100%	89%
Northwestern New Territories						
River Indus	100%	75%	72%	92%	100%	88%
River Beas	94%	58%	81%	100%	100%	87%
River Ganges	100%	47%	58%	94%	100%	80%
Yuen Long Creek	98%	4%	23%	56%	25%	41%
Kam Tin River	100%	0%	29%	67%	50%	49%
Tin Shui Wai Nullah	96%	83%	96%	96%	100%	94%
Fairview Park Nullah	100%	33%	83%	83%	0%	60%
Ha Pak Nai Stream	100%	100%	100%	100%	100%	100%
Tai Shui Hang Stream	100%	100%	100%	100%	100%	100%
Pak Nai Stream	100%	100%	100%	100%	100%	100%
Sheung Pak Nai Stream	100%	92%	100%	100%	100%	98%
Ngau Hom Sha Stream	100%	100%	100%	100%	100%	100%
Tsang Kok Stream	100%	100%	100%	100%	100%	100%
Lantau Island						
Mui Wo River	100%	100%	100%	100%	100%	100%
Tung Chung River	100%	83%	100%	100%	100%	97%
Southwestern New Territories and Kowloon						
Tuen Mun River	100%	65%	93%	85%	100%	89%
Pai Min Kok Stream	92%	96%	100%	100%	100%	98%
Kau Wa Keng Stream	100%	92%	100%	100%	100%	98%
Sam Dip Tam Stream	100%	100%	100%	100%	100%	100%
Kai Tak River	100%	68%	74%	97%	100%	88%
Average Compliance (All monitoring stations)	97%	78%	88%	94%	94%	90%

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Water Quality Index (WQI) for rivers and streams in Hong Kong

Water Quality Index (WQI) is a numerical value used to gauge and denote, based on monitoring data, the overall state of health of rivers and streams. It is relevant to conserving the primary beneficial use of rivers and streams for supporting aquatic life. WQI is calculated based on three key water quality parameters: dissolved oxygen, BOD₅ and ammonia-nitrogen levels. The measurements of these parameters are evaluated using the calculation method shown in the table below:

Calculation of WQI

No. of points awarded	Dissolved Oxygen (% saturation)	BOD ₅ (mg/L)	Ammonia-Nitrogen (mg/L)
1	91 – 110	< 3	< 0.5
2	71 – 90 111 – 120	3.1 – 6.0	0.5 – 1.0
3	51 – 70 121 – 130	6.1 – 9.0	1.1 – 2.0
4	31 – 50	9.1 – 15.0	2.1 – 5.0
5	< 30 or > 130	> 15.0	> 5.0

All parameters carry equal weighting, and the monthly WQI is the sum of points converted from the three measured parameters. The annual WQI of a river monitoring station is the average of all monthly WQI values recorded in the year. The WQI values range from 3 to 15, corresponding to the following gradings of river water quality conditions:

WQI gradings

WQI	River water quality condition
3.0 – 4.5	Excellent
4.6 – 7.5	Good
7.6 – 10.5	Fair
10.6 – 13.5	Bad
13.6 – 15.0	Very Bad

Summary of water quality improvements and mitigation measures in various districts

District	Water quality improvements and mitigation measures	Remaining Issues
New Territories		
Kwai Tsing	All industrial discharges have been controlled under the Water Pollution Control Ordinance (WPCO). Discharges from domestic and commercial premises are connected to public sewers. Water quality in Kau Wa Keng Stream improved from “Bad” in 1991 to “Excellent” in 2019. Its <i>E. coli</i> level has been reduced by over 90% compared with 1998. Sewer provisions for villages are being planned.	Still affected by discharges from a small number of unsewered villages and expedient connections.
Tsuen Wan	All industrial discharges have been controlled under the WPCO. Discharges from domestic and commercial premises are connected to public sewers. Livestock farm discharges in the district have been eliminated under the Livestock Waste Control Scheme (LWCS). The water quality of Sam Dip Tam Stream improved from “Fair” to “Good” in 1988 to “Excellent” in 2019, with 90% reduction in <i>E. coli</i> level as compared with 1999. The WQI grading for Pai Min Kok Stream improved from “Bad” in 1988 to “Excellent” in 2019, with over 90% reduction in <i>E. coli</i> level as compared with 1989. Sewerage works at hinterland of affected beaches have been completed and progressive connection of unsewered premises to new public sewers is being made. Sewer provisions for other villages in the area are being planned.	Still affected by discharges from a small number of unsewered villages and expedient connections.
Tuen Mun	All industrial discharges have been controlled under the WPCO. Discharges from domestic and commercial premises are connected to public sewers. Livestock farm discharges in the district have been controlled under the LWCS. Construction of a trunk sewer to the west of Tuen Mun River was completed in 2015. The WQI gradings for the upper Tuen Mun River improved from “Very Bad” in 1988 to “Bad” to “Good” in 2019, with over 90% reduction in <i>E. coli</i> level. The WQI gradings for the middle and lower river sections substantially improved from “Bad” in 1988 to “Good” in 2019, with over 90% reduction in <i>E. coli</i> level. Sewerage works for villages are being planned and progressively implemented.	Still affected by discharges from unsewered villages especially in the upstream areas and expedient connections.
Yuen Long	Industrial discharges are controlled under the WPCO. Discharges from most of the domestic and commercial premises in the Yuen Long town are connected to public sewers. Under the LWCS, most of the livestock farms in the district had ceased operation. The WQI gradings for upstream sections of Yuen Long Creek and Kam Tin River both changed from “Very Bad” in 1986 to “Bad” to “Fair” in 2019. The <i>E. coli</i> level in these rivers has been reduced by around 80% as compared with 1998. The water quality of Tin Shui Wai Nullah improved from “Fair” to “Bad” in 1993 to “Fair” to “Good” in 2019, with over 90% reduction in <i>E. coli</i> level as compared with 1992. Trunk sewers, pumping stations, and village sewerage are being planned and progressively constructed. The sewage treatment works at San Wai and Yuen Long will be expanded in anticipation of projected population growth, and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial premises and livestock farms.

District	Water quality improvement and mitigation measures	Remaining Issues
New Territories		
North District	Industrial discharges are controlled under the WPCO. Discharges from most of the domestic and commercial premises in the urbanised town areas are connected to public sewers. Under the LWCS, most livestock farms in North District have ceased operation. The WQI gradings for the downstream sections of River Indus, River Beas and River Ganges improved from “Very Bad” in 1987 to “Fair” to “Good” in 2019, with over 90% reduction in <i>E. coli</i> level as compared with 1990. Sewerage works for villages are being planned and progressively implemented. The sewage treatment works at Shek Wu Hui and Sha Tau Kok will be expanded to cater for projected population growth and establishment of the New Development Areas, and upgraded to produce higher quality effluent.	Still affected by polluted discharges from unsewered villages and expedient discharges from industrial premises and livestock farms.
Tai Po	Industrial discharges are controlled under the WPCO. Discharges from most of the domestic and commercial premises in urbanised town areas are connected to public sewers. Under the LWCS, all livestock farms in the district had ceased operation. The water quality of downstream sections of rivers in the area improved from involving “Very Bad” in 1986 to “Good” to “Excellent” in 2019. Also the <i>E. coli</i> level of the downstream sections of both Lam Tsuen River and Tai Po River has been reduced by over 70% compared with 1999. Sewerage works for villages are being planned and progressively implemented. The Tai Po Sewage Treatment Works acquired UV disinfection facilities in 2010, with treatment capacity further expanded to 120000 m ³ /day since 2015.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sha Tin	Industrial discharges are controlled under the WPCO. Discharges from most of the domestic and commercial premises in urbanised town areas are connected to public sewers. Under the LWCS, livestock farms in the district had ceased operation. The water quality of the Shing Mun River main channel improved from “Fair” in 1986 to “Good” in 2019, with over 90% reduction in <i>E. coli</i> level as compared with 1988. Village sewerage is being extended to most of the remaining unsewered villages in the district.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.
Sai Kung	Industrial discharges have been controlled under the WPCO. Discharges from most of the domestic and commercial premises in the urbanised town areas are connected to public sewers. The rural villages houses are being connected to government sewers. Livestock farm discharges in the district have been controlled under the LWCS. The river water quality in the area has improved from largely “Bad” in 1991 to largely “Excellent” in 2019. The <i>E. coli</i> level in rivers in the Port Shelter WCZ has been reduced by over 70% as compared with 1997. Sewerage works for villages are being planned and progressively implemented. Upgrading of the Sai Kung Sewage Treatment Works is under planning.	Still affected by polluted discharges from a small number of unsewered villages and expedient connections.

District	Water quality improvements and mitigation measures	Remaining Issues
New Territories		
Outlying Islands	Discharges from some domestic premises are connected to public sewers. Under the LWCS, all livestock farms in the area had been closed. Mui Wo River and Tung Chung River have maintained “Good” or “Excellent” WQI gradings throughout recent decades. The <i>E. coli</i> level in Mui Wo River has been reduced by over 90% as compared with 1988. Provision and upgrading of village sewerage and sewage treatment facilities are being planned and progressively implemented for Southern Lantau Island, Tung Chung West, Mui Wo, Tai O, Lamma Island, Cheung Chau and Peng Chau.	Still affected by discharges from a small number of unsewered villages and expedient connections.
Kowloon		
Yau Tsim Mong	The urban area of Kowloon contains a few natural watercourses. Industrial and trade discharges have been controlled under the WPCO. Discharges from domestic and commercial premises are connected to public sewers. A large number of expedient connections have been rectified. The water quality of Kai Tak River improved from mostly “Very Bad” in 1986 to all “Good” in 2019. The <i>E. coli</i> level in rivers in the districts has been reduced by over 90% as compared with 1999. The Project of Upgrading of Central and East Kowloon Sewerage Phase 3 commenced in January 2019 to increase the sewerage capacity in Central Kowloon.	Mainly affected by urban surface runoffs, expedient connections and rear lane washing activities.
Sham Shui Po		
Kowloon City		
Wong Tai Sin		
Kwun Tong		
Hong Kong Island		
Central and Western	The urban area of Hong Kong Island contains a few small natural watercourses. Industrial and trade discharges have been controlled under the WPCO. Discharges from domestic and commercial premises are connected to public sewers. A large number of expedient connections have been rectified.	Mainly affected by urban surface runoffs, expedient connections and rear lane washing activities. Some streams are affected by discharges from squatters built near the streams.
Wan Chai		
Eastern		
Southern		